2.5 to 9600SPS, 24-bit Sigma-Delta ADC with PGA and Voltage Reference

FEATURES

Programmable Gain: 1/2/4/8/16/32/64/128/256
Data Rates: 2.5 to 9600SPS
RMS Noise: 30nV at 10SPS (Gain=128)
21.6 Noise Free of Bits at 10SPS (Gain=1)
Offset Drift: 5nV/°C (Gain=128)
Gain Drift: 1ppm/°C
1.17V/2.5V Internal Reference with 5ppm/°C Drift
Integral Non-Linearity: 3ppm
Internal or External Clock
Simultaneous 50Hz/60Hz Rejection
Programmable Current Sources
On-Chip Bias Voltage Generator
Burnout Current Sources
Parity Check
Power Supply
- **AVDD: 2.7V to 5.25V or ±2.5V**
- **DVDD: 2.7V to 5.25V**

Current: 800μA
Package: 16/20/24-lead TSSOP

APPLICATIONS

Weigh Scales
Strain Gauges
Pressure Sensors
Temperature Measurement
Industrial Process Control

DESCRIPTION

The SIG24130/1/2 are low noise, low drift, and high-resolution 24-bit analog-to-digital converters (ADC) with integrated programmable gain amplifier (PGA) and low drift on-chip voltage reference that offers high-accuracy measurement solutions for bridge sensors, thermocouples, and resistance temperature devices (RTD).

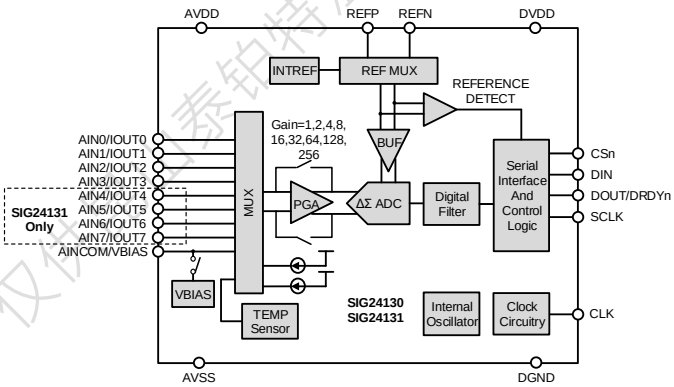
The device contains a low noise PGA with gains selected from 1, 2, 4, 8 16, 32, 64, 128, and 256, a delta-sigma (Δ-Σ) modulator, and a programmable SINC1/SINC4 digital filter. A low drift 1.17V or 2.5V reference is integrated on chip and two matched excitation current sources (IDACs) are provided for accurate RTD measurement. The output data rate from the device can be configured to 2.5 to 9600SPS. Burnout current sources are provided at the analog inputs for sensor connection diagnosis.

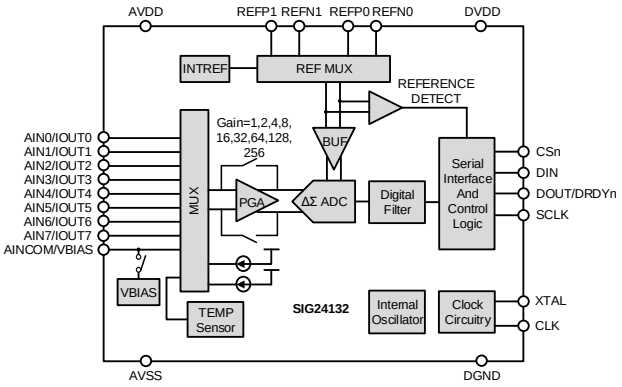
Offset and gain calibration registers are provided with calibration command or direct register write to calibrate the ADC errors or overall system errors. SPI-compatible interface is used for device configuration and parity check is provided for data integrity.

The on-chip oscillator or an external clock can be used as the clock source to the device.

The device can operate with bipolar ±1.35V to ±2.625V analog power supplies, or with a single 2.7V to 5.25V analog power supply.

The SIG24130 is available in 16-lead TSSOP package, the SIG24131 is available in 20-lead TSSOP package, and the SIG24132 is available in 24-lead TSSOP package. All three devices are fully specified over the -40°C to +125°C temperature range.



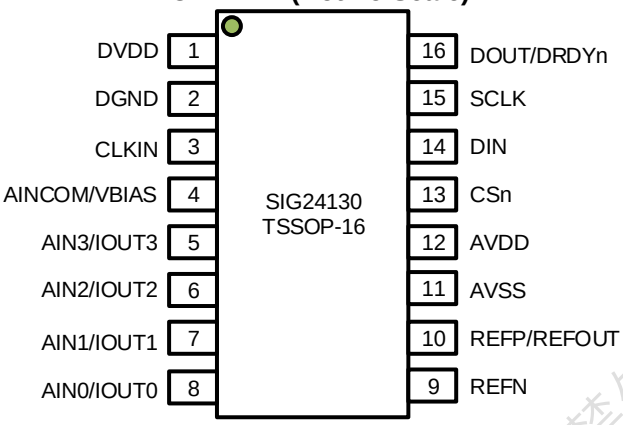


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PIN CONFIGURATION and DESCRIPTIONS

SIG24130

TOP VIEW (Not To Scale)

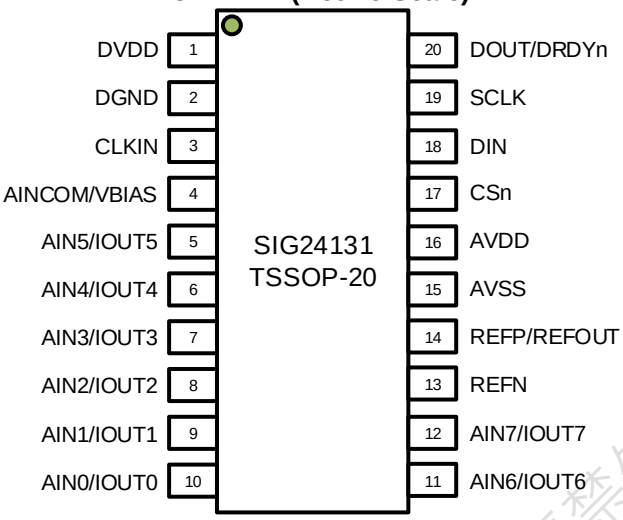


PIN		FUNCTION	DESCRIPTION
NO.	NAME		
1	DVDD	Digital	Digital power supply, 2.7V to 5.25V. DVDD is independent of AVDD.
2	DGND	Digital	Digital ground reference point.
3	CLKIN	Digital Input	1) Internal oscillator: Connect to DGND. 2) External clock: Connect to external clock input.
4	AINCOM/VBIAS	Analog Input	Analog input common, VBIAS.
5	AIN3/IOUT3	Analog Input	Analog input 3, IDAC3.
6	AIN2/IOUT2	Analog Input	Analog input 2, IDAC2.
7	AIN1/IOUT1	Analog Input	Analog input 1, IDAC1.
8	AIN0/IOUT0	Analog Input	Analog input 0, IDAC0.
9	REFN	Analog Input	Negative reference input.
10	REFP/REFOUT	Analog Input	Positive reference input or internal reference output.
11	AVSS	Analog	Negative analog power supply. AVSS can be taken below DVSS to provide bipolar power supplies. For example, AVSS can be tied to –2.5V and AVDD can be tied to +2.5V, providing a ±2.5V dual supplies to the ADC.
12	AVDD	Analog	Positive analog power supply. 2.7V to 5.25V relative to AVSS. AVDD is independent of DVDD.
13	CSn	Digital Input	Serial chip select. Active low.
14	DIN	Digital Input	Serial data input.
15	SCLK	Digital Input	Serial data clock.
16	DOUT/DRDYn	Digital Output	Serial data output and data ready indicator.

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SIG24131

TOP VIEW (Not To Scale)

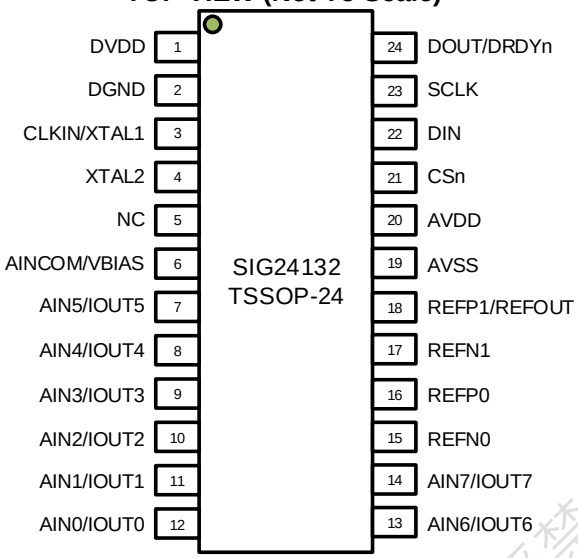


PIN		FUNCTION	DESCRIPTION
NO.	NAME		
1	DVDD	Digital	Digital power supply, 2.7V to 5.25V. DVDD is independent of AVDD.
2	DGND	Digital	Digital ground reference point.
3	CLKIN	Digital Input	1) Internal oscillator: Connect to DGND. 2) External clock: Connect to external clock input.
4	AINCOM/VBIAS	Analog Input	Analog input common, VBIAS.
5	AIN5/IOUT5	Analog Input	Analog input 5, IDAC5.
6	AIN4/IOUT4	Analog Input	Analog input 4, IDAC4.
7	AIN3/IOUT3	Analog Input	Analog input 3, IDAC3.
8	AIN2/IOUT2	Analog Input	Analog input 2, IDAC2.
9	AIN1/IOUT1	Analog Input	Analog input 1, IDAC1.
10	AIN0/IOUT0	Analog Input	Analog input 0, IDAC0.
11	AIN6/IOUT6	Analog Input	Analog input 6, IDAC6.
12	AIN7/IOUT7	Analog Input	Analog input 7, IDAC7.
13	REFN	Analog Input	Negative reference input.
14	REFP/REFOUT	Analog Input	Positive reference input or internal reference output.
15	AVSS	Analog	Negative analog power supply. AVSS can be taken below DVSS to provide bipolar power supplies. For example, AVSS can be tied to –2.5V and AVDD can be tied to +2.5V, providing a ±2.5V dual supplies to the ADC.
16	AVDD	Analog	Positive analog power supply. 2.7V to 5.25V relative to AVSS. AVDD is independent of DVDD.
17	CSn	Digital Input	Serial chip select. Active low.
18	DIN	Digital Input	Serial data input.
19	SCLK	Digital Input	Serial data clock.
20	DOUT/DRDYn	Digital Output	Serial data output and data ready indicator.

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SIG24132

TOP VIEW (Not To Scale)



PIN		FUNCTION	DESCRIPTION
NO.	NAME		
1	DVDD	Digital	Digital power supply, 2.7V to 5.25V. DVDD is independent of AVDD.
2	DGND	Digital	Digital ground reference point.
3	CLKIN/XTAL1	Digital Input	1) Internal oscillator: Connect to DGND. 2) External clock: Connect to external clock input. 3) Crystal oscillator connection 1.
4	XTAL2	Digital Input	Crystal oscillator connection 2.
5	NC	Digital	No connection (float) or connect to DVDD/DGND.
6	AINCOM/VBIAS	Analog Input	Analog input common, VBIAS.
7	AIN5/IOUT5	Analog Input	Analog input 5, IDAC5.
8	AIN4/IOUT4	Analog Input	Analog input 4, IDAC4.
9	AIN3/IOUT3	Analog Input	Analog input 3, IDAC3.
10	AIN2/IOUT2	Analog Input	Analog input 2, IDAC2.
11	AIN1/IOUT1	Analog Input	Analog input 1, IDAC1.
12	AIN0/IOUT0	Analog Input	Analog input 0, IDAC0.
13	AIN6/IOUT6	Analog Input	Analog input 6, IDAC6.
14	AIN7/IOUT7	Analog Input	Analog input 7, IDAC7.
15	REFN0	Analog Input	Negative reference input 0.
16	REFP0	Analog Input	Positive reference input 0.
17	REFN1	Analog Input	Negative reference input 1.
18	REFP1/REFOUT	Analog Input	Positive reference input 1 or internal reference output.
19	AVSS	Analog	Negative analog power supply. AVSS can be taken below DVSS to provide bipolar power supplies. For example, AVSS can be tied to –2.5V and AVDD can be tied to +2.5V, providing a ±2.5V dual supplies to the ADC.
20	AVDD	Analog	Positive analog power supply. 2.7V to 5.25V relative to AVSS. AVDD is independent of DVDD.
21	CSn	Digital Input	Serial chip select. Active low.
22	DIN	Digital Input	Serial data input.
23	SCLK	Digital Input	Serial data clock.
24	DOUT/DRDYn	Digital Output	Serial data output and data ready indicator.

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PACKAGE/ORDERING INFORMATION

MODEL	PACKAGE DESCRIPTION	SPECIFIED TEMPERATURE RANGE	ORDERING NUMBER	PACKING OPTION
SIG24130	TSSOP-16	-40°C to +125°C	SIG24130-ITSP16-RL	Reel, 5000
SIG24131	TSSOP-20	-40°C to +125°C	SIG24131-ITSP20-RL	Reel, 4500
SIG24132	TSSOP-24	-40°C to +125°C	SIG24132-ITSP24-RL	Reel, 3000

SPECIFICATIONS

Absolute Maximum Ratings

Over operating free-air temperature range, unless otherwise noted.(1)

		MIN	MAX	UNIT
Voltage	AVDD to AVSS	–0.3	6.5	V
	AVSS to DGND	–3	0.3	V
	DVDD to DGND	–0.3	6.5	V
	Analog input	V_{AVSS} – 0.3	V_{AVDD} + 0.3	V
	Digital input	V_{DGND} – 0.3	V_{DVDD} + 0.3	V
Current	Input current	–10	10	mA
Temperature	Junction (T_J)	–50	150	°C
	Storage (T_{stg})	-60	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These is a stress rating only; functional operation of the device at these or any other conditions above those indicated in the operational section of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

ESD Ratings

SYMBOL	PARAMTER	CONDITION	VALUE	UNIT
HBM	Human-body Model	ANSI/ESDA/JEDEC JS-001	±4000	V
CDM	Charged-device model	JEDEC EIA/JS-002-2022	±2000	V

This integrated circuit can be damaged by ESD. Signal Micro recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage. ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

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Electrical Characteristics

Minimum/Maximum specifications apply from -40˚C to +125˚C. Typical specifications are at +25˚C. All specification are at V_{AVDD}=5V, V_{AVSS}=0V, V_{DVDD}=3.3V, V_{REF}=2.5V, f_{CLK}=1.536MHz, data rate=10SPS, and PGA Gain=1, unless otherwise noted.

PARAMETER	TEST CONDITION OR NOTES	MIN(1)	TYP	MAX(1)	UNITS
ANALOG INPUTS					
Differential Input Voltage	$V_{IN} = V_{INP} – V_{INN}$	$–V_{REF}$/Gain		$+V_{REF}$/Gain	V
Absolute Input Voltage	PGA bypass	V_{AVSS} – 0.05		V_{AVDD} + 0.05	V
	PGA enabled	V_{AVSS} + 0.2		V_{AVDD} – 0.2	
Common MODE Input Range	PGA enabled	V_{AVSS} + 0.2 + $\lvert V_{INMAX}\rvert$·Gain/2		V_{AVDD} – 0.2 – $\lvert V_{INMAX}\rvert$·Gain/2	V
Absolute Input Current	PGA bypass		±10		nA
	PGA enabled		±1		nA
SYSTEM PERFORMANCE					
PGA Gain			1/2/4/8/16/32/ 64/128/256		V/V
Resolution			24		Bits
Data Rate		2.5		9600	SPS
Noise		See Noise Table			
Integral Nonlinearity (INL)			±3		ppm
Offset Error	All PGA gains		±256/Gain		µV
	After calibration		In order of noise		
Offset Drift vs. Temperature	All PGA gains		±256/Gain ± 3		nV/°C
Gain Error(2)	Gain=1 to 128	-300	±100	300	ppm
	Gain=256	-450	±200	450	ppm
Gain Mismatch(2)	Gain=1 to 128		200	350	ppm
Gain Drift vs. Temperature	All PGA gains	-3	±1	3	ppm/°C
Normal MODE Rejection (NMRR)	f_{IN}=50/60Hz, ±2%, data rate=10SPS		See Table 1		dB
Common MODE Rejection (CMRR)	f_{IN}=50/60Hz, data rate=1200SPS	100	120		dB
Power Supply Rejection (PSRR)	AVDD, AVSS	75	90		dB
	DVDD	80	120		dB
EXTERNAL REFERENCE INPUT					
Differential Reference Voltage (V_{REF})	$V_{REF} = V_{REFP} – V_{REFN}$	0.5		$V_{AVDD} – V_{AVSS}$ + 0.1	V
Absolute Negative Reference Voltage (V_{REFN})		V_{AVSS} – 0.05		V_{REFP} – 0.5	V
Absolute Positive Reference Voltage (V_{REFP})		V_{REFN} + 0.5		V_{AVDD} + 0.05	V
Average Voltage Input Current			50		nA
INTERNAL VOLTAGE REFERENCE					
Reference Voltage			2.5		V
Initial Accuracy	T_A = 25˚C	-0.1%	±0.01%	+0.1%	
Voltage Temperature Drift	T_A = -40˚C to 125˚C	-20	±5	+20	ppm/°C
Power Supply Rejection			90		dB
EXCITATION CURRENT SOURCES					
Output Current	V_{REFP} = 2.5V	10/50/100/200/250/500/1000			µA
Compliance Voltage(3)	IDAC <= 250µA	V_{AVSS}		V_{AVDD} – 0.5	V
	IDAC = 500µA	V_{AVSS}		V_{AVDD} – 0.6	
	IDAC = 1000µA	V_{AVSS}		V_{AVDD} – 0.8	
Accuracy	IDAC =10µA	-4%	±2%	+4%	
	IDAC >= 50µA	-2%	±1%	+2%	
Current Mismatch	IDAC = 10µA	-3.0%	±0.60%	+3.0%	

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	IDAC = 50μA	-1.5%	±0.25%	+1.5%	
	IDAC = 100μA	-1.2%	±0.20%	+1.2%	
	IDAC = 200μA, 250μA	-1.0%	±0.18%	+1.0%	
	IDAC = 500μA	-0.8%	±0.12%	+0.8%	
	IDAC = 1000μA	-0.6%	±0.1%	+0.6%	
Temperature Drift		-90	±20	+90	ppm/°C
Temperature Drift Mismatch		-30	±5	+30	ppm/°C
BURNOUT CURRENT SOURCES					
Current Setting			1		μA
TEMPERATURE SENSOR					
Output Voltage			113.3		mV
Temperature Coefficient			377.6		uV/°C
ADC CLOCK					
External Clock	Frequency Range	1	1.536	1.6	MHz
	Duty Cycle	40%		60%	
Internal Oscillator	Nominal Frequency		1.536		MHz
	Accuracy	–2%	±0.5%	+2%	
DIGITAL INPUT/OUTPUT					
High-level Output Voltage (V_{OH})	I_{OH} = 4mA	0.8·V_{DVDD}			V
Low-level Output Voltage (V_{OL})	I_{OL} = –4mA			0.2·V_{DVDD}	V
High-level Input Voltage (V_{IH})		0.7·V_{DVDD}		V_{DVDD}	V
Low-level Input Voltage (V_{IL})		V_{DGND}		0.3·V_{DVDD}	V
Input Hysteresis			0.5		V
Input Leakage				±10	μA
POWER SUPPLY					
AVSS Voltage (V_{AVSS})		–2.625		0	V
AVDD Voltage (V_{AVDD})		V_{AVSS} + 2.7		V_{AVSS} + 5.25	V
DVDD Voltage (V_{DVDD})		2.7		5.25	V
AVDD, AVSS Current (I_{AVDD})	Buffer Off		250	350	μA
	Buffer On		600	750	μA
	Sleep MODE		1		μA
DVDD Current (I_{DVDD})	Active MODE		180	300	μA
	Sleep MODE		40		μA
Total Power Dissipation	Buffer Off		1.9		mW
	Buffer On		3.6		mW
	Sleep MODE		0.13		mW
TEMPERATURE RANGE					
Specified temperature range		–40		125	°C
Operating temperature range		–50		125	°C
Storage temperature range		–60		150	°C

(1) Compliance to datasheet limits is assured by one or more methods: production test, characterization and/or design.
(2) MIN and MAX values listed for gain error are for +25˚C room temperature only.
(3) The IDAC current does not change by more than 0.01% from the nominal value when staying within the specified compliance voltage.

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Timing Requirements: Serial Interface

Over the operating ambient temperature range and DVDD = 2.7V to 5.25V, unless otherwise noted.

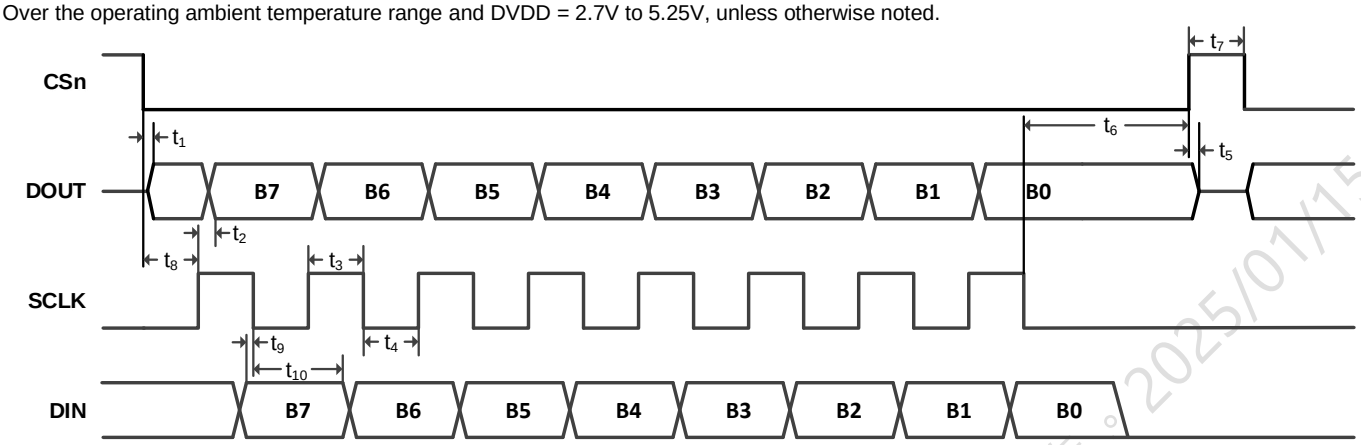


Figure 1. Serial Interface Timing Requirements

SYMBOL	DESCRIPTION	MIN	MAX	UNIT
t_1	CSn falling edge to DOUT/DRDYn driven: propagation delay(1)		50	ns
t_2	SCLK rising edge to valid DOUT/DRDYn: propagation delay(1)		50	ns
t_3	SCLK high pulse width	100		ns
t_4	SCLK low pulse width	100		ns
	SCLK period	200	10^6	ns
t_5	CSn rising edge to DOUT high impedance: propagation delay		40	ns
t_6	Last SCLK falling edge to CSn rising edge: delay time	50		ns
t_7	CSn high pulse width	50		ns
t_8	CSn falling edge to first SCLK rising edge: setup time(2)	50		ns
t_9	Valid DIN to SCLK falling edge: setup time	50		ns
t_{10}	Valid DIN to SCLK falling edge: hold time	25		ns

(1) DOUT load = 20pF || 100kΩ to DGND.
(2) CSn can be tied low.

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NOISE PERFORMANCE

The noise performance of the ADC is affected by PGA gain, data rate, and digital filter setting. The following tables show the rms noise and peak-to-peak noise for SINC4 and SINC1 filters. The effective number of bits (ENOB) and noise-free bits are also listed according to Equation (1) and Equation (2):

$$\text{ENOB} = \log_2(2 \times V_{REF} / \text{Gain} / V_{RMS}) \quad (1)$$

$$\text{Noise Free Bits} = \log_2(2 \times V_{REF} / \text{Gain} / V_{p\text{-}p}) \quad (2)$$

The noise data listed in the table are typical and are generated from continuous ADC readings with differential input voltage of 0 V.

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Table 1. ADC Noise in µVRMS (µVPP) at T_A = 25°C, V_{AVDD} = 5 V, V_{AVSS} = 0 V, V_{REF} = 2.5 V, SINC4 Filter

Data Rate (SPS)	PGA GAIN								
	1	2	4	8	16	32	64	128	256
2.5	0.280(1.53)	0.160(0.833)	0.098(0.471)	0.052(0.267)	0.035(0.192)	0.023(0.105)	0.017(0.088)	0.015(0.077)	0.014(0.066)
5	0.396(2.16)	0.227(1.18)	0.138(0.667)	0.074(0.378)	0.049(0.271)	0.032(0.148)	0.025(0.125)	0.021(0.109)	0.019(0.093)
10	0.559(3.05)	0.320(1.67)	0.196(0.943)	0.105(0.535)	0.070(0.383)	0.046(0.209)	0.035(0.177)	0.030(0.155)	0.028(0.132)
12.5	0.625(3.41)	0.358(1.86)	0.219(1.05)	0.117(0.598)	0.078(0.428)	0.051(0.234)	0.039(0.198)	0.033(0.173)	0.031(0.147)
15	0.685(3.74)	0.392(2.04)	0.240(1.15)	0.128(0.655)	0.085(0.469)	0.056(0.256)	0.043(0.217)	0.036(0.189)	0.034(0.161)
20	0.791(4.31)	0.453(2.36)	0.277(1.33)	0.148(0.756)	0.098(0.542)	0.064(0.296)	0.049(0.250)	0.042(0.219)	0.039(0.186)
25	0.884(4.82)	0.507(2.64)	0.309(1.49)	0.166(0.845)	0.110(0.606)	0.072(0.331)	0.055(0.280)	0.047(0.245)	0.044(0.208)
30	0.969(5.28)	0.555(2.89)	0.339(1.63)	0.182(0.926)	0.121(0.663)	0.079(0.362)	0.060(0.306)	0.051(0.268)	0.048(0.228)
40	1.12(6.10)	0.641(3.33)	0.391(1.89)	0.210(1.07)	0.139(0.766)	0.091(0.418)	0.070(0.354)	0.059(0.309)	0.055(0.263)
50	1.25(6.82)	0.717(3.73)	0.438(2.11)	0.235(1.20)	0.156(0.856)	0.102(0.468)	0.078(0.395)	0.066(0.346)	0.062(0.294)
60	1.37(7.47)	0.785(4.08)	0.479(2.31)	0.257(1.31)	0.171(0.938)	0.112(0.512)	0.085(0.433)	0.072(0.379)	0.067(0.322)
80	1.58(8.63)	0.906(4.71)	0.554(2.67)	0.297(1.51)	0.197(1.08)	0.129(0.592)	0.098(0.500)	0.084(0.438)	0.078(0.372)
100	1.77(9.65)	1.01(5.27)	0.619(2.98)	0.332(1.69)	0.220(1.21)	0.144(0.662)	0.110(0.559)	0.093(0.489)	0.087(0.416)
120	2.03(10.7)	1.24(7.31)	0.713(3.47)	0.347(1.94)	0.214(1.17)	0.156(0.932)	0.103(0.547)	0.102(0.566)	0.088(0.462)
160	2.16(10.4)	1.40(8.65)	0.739(4.40)	0.446(2.42)	0.271(1.27)	0.175(0.969)	0.130(0.622)	0.108(0.615)	0.105(0.511)
200	2.48(14.9)	1.65(9.65)	0.856(4.40)	0.466(2.67)	0.306(1.41)	0.177(1.12)	0.126(0.664)	0.119(0.705)	0.125(0.701)
240	3.19(16.4)	1.68(10.6)	0.867(4.92)	0.530(2.94)	0.297(1.62)	0.218(1.37)	0.141(0.759)	0.146(0.770)	0.132(0.669)
250	3.18(15.8)	2.01(10.8)	1.08(6.30)	0.641(3.15)	0.393(2.38)	0.249(1.34)	0.149(0.799)	0.152(0.865)	0.134(0.706)
300	3.42(16.7)	2.05(11.5)	1.03(6.75)	0.586(3.32)	0.379(2.34)	0.250(1.47)	0.182(1.10)	0.154(0.889)	0.160(0.850)
400	3.40(19.1)	2.25(13.0)	1.22(7.79)	0.692(3.86)	0.430(2.63)	0.257(1.55)	0.217(1.17)	0.167(1.09)	0.162(1.25)
480	4.37(33.2)	2.54(15.5)	1.36(8.54)	0.730(4.53)	0.455(2.53)	0.304(1.77)	0.232(1.60)	0.188(1.02)	0.184(1.14)
500	4.12(26.1)	2.75(19.9)	1.51(8.91)	0.852(5.80)	0.501(2.82)	0.340(2.29)	0.251(1.42)	0.207(1.19)	0.189(1.04)
600	4.49(27.3)	2.81(16.4)	1.62(8.76)	0.872(5.81)	0.500(2.93)	0.339(2.21)	0.250(1.47)	0.209(1.31)	0.212(1.36)
800	5.29(34.7)	3.12(22.0)	1.76(12.6)	0.953(7.12)	0.604(4.10)	0.384(2.79)	0.307(1.77)	0.246(1.92)	0.245(1.45)
1000	6.66(46.7)	3.87(25.2)	2.16(14.0)	1.18(7.04)	0.690(4.53)	0.477(3.33)	0.349(2.15)	0.287(1.96)	0.270(1.74)
1200	6.36(44.0)	3.87(23.1)	2.14(14.5)	1.15(7.27)	0.729(4.83)	0.493(3.44)	0.354(2.66)	0.326(2.27)	0.303(1.95)
2000	9.53(68.6)	5.76(39.7)	3.10(23.6)	1.73(10.2)	1.05(6.70)	0.699(4.71)	0.497(3.89)	0.423(2.94)	0.390(2.93)
2400	9.61(58.1)	5.55(34.9)	3.11(21.9)	1.72(12.9)	1.00(6.52)	0.684(4.78)	0.511(3.17)	0.426(3.17)	0.421(3.21)
4000	30.4(213)	16.4(120)	8.16(55.7)	4.20(28.2)	2.27(16.2)	1.29(9.73)	0.842(6.37)	0.649(4.77)	0.565(4.17)
4800	30.9(223)	16.3(122)	8.20(64.2)	4.27(30.8)	2.29(17.2)	1.29(9.36)	0.843(5.90)	0.664(5.00)	0.588(4.82)
8000	311(3520)	154(1140)	79.1(730)	38.7(380)	19.2(163)	9.88(82.2)	4.94(37.4)	2.56(18.0)	1.44(11.2)
9600	307(2690)	156(1370)	77.9(623)	39.1(368)	19.9(180)	9.79(88.5)	5.14(41.9)	2.53(17.6)	1.48(11.1)

Table 2. ADC ENOB (Noise Free Bits) at T_A = 25°C, V_{AVDD} = 5 V, V_{AVSS} = 0 V, V_{REF} = 5 V, SINC4 Filter

Data Rate (SPS)	PGA GAIN								
	1	2	4	8	16	32	64	128	256
2.5	25.1(22.6)	24.7(22.3)	24.6(22.2)	24.5(22.1)	24.1(21.7)	23.7(21.2)	23.2(20.9)	22.4(20.1)	21.5(19.1)
5	24.6(22.1)	24.2(21.8)	24.1(21.7)	24.0(21.6)	23.6(21.2)	23.2(20.7)	22.7(20.4)	21.9(19.6)	21.0(18.6)
10	24.1(21.6)	23.7(21.3)	23.6(21.2)	23.5(21.1)	23.1(20.7)	22.7(20.2)	22.2(19.9)	21.4(19.1)	20.5(18.1)
12.5	23.9(21.5)	23.5(21.2)	23.5(21.0)	23.4(21.0)	23.0(20.5)	22.5(20.0)	22.0(19.7)	21.2(18.9)	20.3(17.9)
15	23.8(21.3)	23.4(21.0)	23.3(20.9)	23.3(20.8)	22.8(20.4)	22.4(19.9)	21.9(19.6)	21.1(18.8)	20.2(17.8)
20	23.6(21.1)	23.2(20.8)	23.1(20.7)	23.0(20.6)	22.6(20.2)	22.2(19.7)	21.7(19.4)	20.9(18.6)	20.0(17.6)
25	23.4(21.0)	23.0(20.7)	23.0(20.5)	22.9(20.5)	22.5(20.0)	22.0(19.5)	21.5(19.2)	20.7(18.4)	19.8(17.4)
30	23.3(20.8)	22.9(20.5)	22.8(20.4)	22.8(20.3)	22.3(19.9)	21.9(19.4)	21.4(19.1)	20.6(18.3)	19.7(17.3)
40	23.1(20.6)	22.7(20.3)	22.6(20.2)	22.5(20.1)	22.1(19.7)	21.7(19.2)	21.2(18.9)	20.4(18.1)	19.5(17.1)
50	22.9(20.5)	22.5(20.2)	22.5(20.0)	22.4(20.0)	22.0(19.5)	21.5(19.0)	21.0(18.7)	20.2(17.9)	19.3(16.9)
60	22.8(20.3)	22.4(20.0)	22.3(19.9)	22.3(19.8)	21.8(19.4)	21.4(18.9)	20.9(18.6)	20.1(17.8)	19.2(16.8)
80	22.6(20.1)	22.2(19.8)	22.1(19.7)	22.0(19.6)	21.6(19.2)	21.2(18.7)	20.7(18.4)	19.9(17.6)	19.0(16.6)
100	22.4(20.0)	22.0(19.7)	22.0(19.5)	21.9(19.5)	21.5(19.0)	21.0(18.5)	20.5(18.2)	19.7(17.4)	18.8(16.4)
120	22.2(19.6)	21.9(19.4)	22.0(19.9)	21.6(19.4)	21.4(18.9)	21.1(18.8)	20.4(18.3)	19.6(17.1)	18.5(16.0)
160	22.1(19.4)	21.7(19.2)	21.5(18.8)	21.4(19.0)	21.2(18.8)	20.8(18.3)	20.2(17.6)	19.3(17.0)	18.5(16.1)
200	21.9(19.5)	21.5(19.3)	21.4(19.0)	21.2(19.0)	21.0(18.6)	20.7(18.1)	20.0(17.3)	19.3(16.9)	18.5(15.9)
240	21.7(19.4)	21.5(19.1)	21.4(18.8)	21.2(18.5)	20.9(18.5)	20.4(17.9)	19.8(17.2)	19.1(16.6)	18.1(15.7)
250	21.6(19.2)	21.3(18.9)	21.1(18.6)	20.9(18.4)	20.7(18.1)	20.3(17.7)	19.8(17.3)	19.0(16.1)	18.1(15.6)
300	21.5(19.2)	21.3(18.7)	21.1(18.4)	20.9(18.3)	20.8(18.5)	20.3(17.8)	19.8(17.4)	19.0(16.4)	18.1(15.8)
400	21.4(18.8)	21.0(18.3)	20.9(18.3)	20.7(18.1)	20.6(18.0)	20.1(17.3)	19.6(17.0)	18.8(16.2)	17.8(15.3)
480	21.3(18.6)	20.9(18.4)	20.7(18.2)	20.7(18.0)	20.4(18.0)	20.0(17.4)	19.4(16.8)	18.6(15.8)	17.6(15.3)
500	21.0(18.3)	20.7(18.2)	20.8(18.2)	20.4(17.7)	20.2(17.6)	19.9(17.4)	19.2(16.5)	18.6(16.1)	16.2(12.3)
600	21.1(18.4)	20.8(18.0)	20.7(18.0)	20.6(17.8)	20.2(17.7)	19.7(17.1)	19.3(16.7)	18.5(15.9)	17.6(15.1)
800	20.9(18.3)	20.6(17.9)	20.4(17.8)	20.3(17.3)	20.0(17.2)	19.6(17.0)	19.1(16.3)	18.2(15.5)	17.3(14.6)
1000	20.6(18.1)	20.2(17.6)	20.2(17.3)	20.0(17.5)	19.8(16.9)	19.3(16.3)	18.7(16.1)	18.0(15.1)	17.1(14.3)
1200	20.6(17.9)	20.2(17.5)	20.2(17.6)	20.0(17.2)	19.8(17.1)	19.4(16.8)	18.8(16.1)	18.0(15.3)	17.0(14.3)
2000	19.9(17.2)	19.7(16.8)	19.5(16.8)	19.4(16.6)	19.2(16.4)	18.7(16.1)	18.2(15.4)	17.5(14.8)	16.6(13.9)
2400	19.8(17.0)	19.6(16.7)	19.5(16.7)	19.4(16.5)	19.2(16.4)	18.8(16.0)	18.2(15.5)	17.4(14.6)	16.5(13.8)
4000	17.4(14.5)	17.4(14.3)	17.3(14.3)	17.4(14.6)	17.3(14.2)	17.3(14.1)	17.1(14.2)	16.7(13.8)	16.0(13.2)
4800	17.4(14.4)	17.4(14.1)	17.4(14.3)	17.3(14.3)	17.3(14.2)	17.2(14.2)	17.1(14.1)	16.7(13.8)	15.9(13.0)
8000	14.0(11.0)	14.0(11.1)	14.0(10.9)	14.0(10.8)	14.0(10.8)	14.0(10.6)	14.0(10.7)	13.9(10.9)	13.9(11.1)
9600	13.9(10.2)	14.0(10.9)	14.0(10.7)	13.9(10.8)	14.0(10.9)	14.0(11.0)	14.0(10.8)	13.9(11.0)	13.9(10.9)

Table 3. ADC Noise in µVRMS (µVPP) at T_A = 25°C, V_{AVDD} = 5 V, V_{AVSS} = 0 V, V_{REF} = 2.5 V, SINC1 Filter

Data Rate (SPS)	PGA GAIN								
	1	2	4	8	16	32	64	128	256
2.5	0.440(2.50)	0.255(1.41)	0.132(0.707)	0.076(0.365)	0.045(0.227)	0.032(0.141)	0.023(0.134)	0.019(0.098)	0.018(0.095)
5	0.622(3.53)	0.361(2.00)	0.186(1.00)	0.108(0.517)	0.063(0.321)	0.045(0.200)	0.033(0.190)	0.026(0.139)	0.026(0.134)
10	0.880(5.00)	0.511(2.83)	0.263(1.41)	0.153(0.731)	0.089(0.454)	0.063(0.283)	0.046(0.268)	0.037(0.197)	0.036(0.190)
12.5	0.984(5.59)	0.571(3.16)	0.294(1.58)	0.171(0.817)	0.100(0.507)	0.071(0.316)	0.052(0.300)	0.042(0.220)	0.041(0.212)
15	1.08(6.12)	0.626(3.46)	0.322(1.73)	0.187(0.895)	0.109(0.556)	0.077(0.346)	0.057(0.328)	0.046(0.241)	0.045(0.233)
20	1.24(7.07)	0.723(4.00)	0.372(2.00)	0.216(1.03)	0.126(0.642)	0.089(0.400)	0.065(0.379)	0.053(0.278)	0.051(0.269)
25	1.39(7.90)	0.808(4.47)	0.416(2.24)	0.241(1.16)	0.141(0.717)	0.100(0.447)	0.073(0.424)	0.059(0.311)	0.058(0.301)
30	1.52(8.65)	0.885(4.90)	0.456(2.45)	0.265(1.27)	0.154(0.786)	0.109(0.490)	0.080(0.464)	0.064(0.341)	0.063(0.329)
40	1.76(9.99)	1.02(5.66)	0.526(2.83)	0.305(1.46)	0.178(0.907)	0.126(0.566)	0.092(0.536)	0.074(0.393)	0.073(0.380)
50	1.97(11.2)	1.14(6.32)	0.588(3.16)	0.341(1.63)	0.199(1.01)	0.141(0.632)	0.103(0.600)	0.083(0.440)	0.081(0.425)
60	2.16(12.2)	1.25(6.93)	0.645(3.46)	0.374(1.79)	0.218(1.11)	0.155(0.693)	0.113(0.657)	0.091(0.482)	0.089(0.466)
80	2.49(14.1)	1.45(8.00)	0.744(4.00)	0.432(2.07)	0.252(1.28)	0.179(0.800)	0.131(0.758)	0.105(0.556)	0.103(0.538)
100	2.78(15.8)	1.62(8.94)	0.832(4.47)	0.483(2.31)	0.282(1.43)	0.200(0.894)	0.146(0.848)	0.118(0.622)	0.115(0.601)
120	3.02(15.5)	1.76(9.39)	0.926(5.44)	0.570(3.21)	0.304(1.57)	0.210(1.18)	0.172(0.876)	0.129(0.719)	0.131(0.659)
160	3.45(18.0)	2.14(11.2)	1.10(4.92)	0.622(3.32)	0.435(2.34)	0.254(1.42)	0.176(0.908)	0.169(0.821)	0.145(0.941)
200	4.01(20.9)	2.32(12.8)	1.28(8.31)	0.662(3.32)	0.390(2.05)	0.293(1.80)	0.216(1.09)	0.186(1.05)	0.172(1.00)
240	4.34(22.5)	2.49(13.3)	1.33(7.34)	0.757(3.97)	0.485(2.87)	0.303(1.65)	0.234(1.32)	0.188(1.03)	0.185(0.958)
250	4.48(24.7)	2.69(14.3)	1.51(7.49)	0.848(4.71)	0.489(3.40)	0.319(1.77)	0.263(1.54)	0.199(1.02)	0.214(1.46)
300	4.55(23.4)	2.61(16.8)	1.41(8.76)	0.823(4.86)	0.489(2.53)	0.314(1.79)	0.242(1.44)	0.226(1.41)	0.219(1.42)
400	4.96(30.6)	3.16(17.7)	1.66(9.73)	0.956(6.37)	0.567(3.07)	0.365(2.48)	0.276(1.43)	0.251(1.63)	0.215(1.29)
480	5.64(39.8)	3.32(19.8)	1.82(9.35)	0.992(5.87)	0.626(3.24)	0.385(2.36)	0.312(1.98)	0.267(1.62)	0.246(1.27)
500	5.99(43.7)	3.67(21.9)	2.00(12.5)	1.04(5.98)	0.684(4.01)	0.463(2.97)	0.339(2.18)	0.269(1.61)	0.259(1.45)
600	6.43(39.8)	4.00(28.0)	2.01(11.3)	1.16(6.71)	0.659(4.30)	0.450(2.66)	0.338(1.86)	0.294(2.04)	0.265(1.79)
800	7.21(51.9)	4.21(29.4)	2.31(13.6)	1.31(8.65)	0.760(4.62)	0.526(3.57)	0.381(2.46)	0.332(1.99)	0.311(1.94)
1000	7.95(53.5)	4.69(30.1)	2.56(15.0)	1.48(10.4)	0.856(5.67)	0.575(3.43)	0.443(2.44)	0.369(2.44)	0.333(2.09)
1200	8.36(54.0)	4.83(36.5)	2.67(19.5)	1.43(10.0)	0.914(6.06)	0.604(3.86)	0.432(2.73)	0.377(2.47)	0.371(2.31)
2000	9.70(59.6)	5.75(38.0)	3.08(21.0)	1.74(11.2)	1.02(6.92)	0.663(4.93)	0.508(3.48)	0.419(2.77)	0.376(2.58)
2400	9.50(68.9)	5.79(37.7)	3.17(25.2)	1.74(12.0)	1.02(7.66)	0.682(4.34)	0.524(3.63)	0.441(3.03)	0.421(2.90)
4000	31.2(216)	15.6(128)	8.16(62.4)	4.23(32.7)	2.29(18.9)	1.30(8.85)	0.849(6.04)	0.635(4.56)	0.556(4.20)
4800	32.0(303)	16.3(136)	8.14(57.2)	4.21(34.3)	2.26(18.0)	1.28(9.36)	0.853(6.55)	0.673(5.32)	0.584(4.56)
8000	309(2690)	158(1320)	78.1(777)	40.0(451)	19.6(200)	9.89(95.6)	4.92(45.6)	2.55(22.0)	1.43(12.0)
9600	311(3540)	157(1660)	78.1(754)	38.7(318)	19.6(213)	9.88(80.8)	4.96(36.3)	2.55(19.5)	1.44(10.9)

Table 4. ADC ENOB (Noise Free Bits) at T_A = 25°C, V_{AVDD} = 5 V, V_{AVSS} = 0 V, V_{REF} = 5 V, SINC1 Filter

Data Rate (SPS)	PGA GAIN								
	1	2	4	8	16	32	64	128	256
2.5	24.5(22.2)	24.0(21.6)	24.1(21.7)	23.8(21.4)	23.6(21.2)	23.2(21.0)	22.7(20.3)	21.9(19.6)	21.0(18.7)
5	24.0(21.7)	23.5(21.1)	23.6(21.2)	23.3(20.9)	23.1(20.7)	22.7(20.5)	22.2(19.8)	21.4(19.1)	20.5(18.2)
10	23.5(21.2)	23.0(20.6)	23.1(20.7)	22.8(20.4)	22.6(20.2)	22.2(20.0)	21.7(19.3)	20.9(18.6)	20.0(17.7)
12.5	23.3(21.0)	22.9(20.4)	22.9(20.5)	22.7(20.3)	22.4(20.1)	22.1(19.8)	21.6(19.1)	20.8(18.4)	19.8(17.5)
15	23.2(20.9)	22.7(20.3)	22.8(20.4)	22.6(20.2)	22.3(20.0)	21.9(19.7)	21.4(19.0)	20.6(18.3)	19.7(17.4)
20	23.0(20.7)	22.5(20.1)	22.6(20.2)	22.3(19.9)	22.1(19.7)	21.7(19.5)	21.2(18.8)	20.4(18.1)	19.5(17.2)
25	22.8(20.5)	22.4(19.9)	22.4(20.0)	22.2(19.8)	21.9(19.6)	21.6(19.3)	21.1(18.6)	20.3(17.9)	19.3(17.0)
30	22.7(20.4)	22.2(19.8)	22.3(19.9)	22.1(19.7)	21.8(19.5)	21.4(19.2)	20.9(18.5)	20.1(17.8)	19.2(16.9)
40	22.5(20.2)	22.0(19.6)	22.1(19.7)	21.8(19.4)	21.6(19.2)	21.2(19.0)	20.7(18.3)	19.9(17.6)	19.0(16.7)
50	22.3(20.0)	21.9(19.4)	21.9(19.5)	21.7(19.3)	21.4(19.1)	21.1(18.8)	20.6(18.1)	19.8(17.4)	18.8(16.5)
60	22.2(19.9)	21.7(19.3)	21.8(19.4)	21.6(19.2)	21.3(19.0)	20.9(18.7)	20.4(18.0)	19.6(17.3)	18.7(16.4)
80	22.0(19.7)	21.5(19.1)	21.6(19.2)	21.3(18.9)	21.1(18.7)	20.7(18.5)	20.2(17.8)	19.4(17.1)	18.5(16.2)
100	21.8(19.5)	21.4(18.9)	21.4(19.0)	21.2(18.8)	20.9(18.6)	20.6(18.3)	20.1(17.6)	19.3(16.9)	18.3(16.0)
120	21.7(19.3)	21.3(18.9)	21.1(18.6)	21.1(18.5)	21.1(18.4)	20.5(18.0)	19.8(17.2)	19.1(16.6)	18.1(15.7)
160	21.4(19.2)	21.2(18.9)	21.0(18.6)	21.0(18.8)	20.6(18.0)	20.4(18.1)	20.0(17.5)	18.8(16.3)	18.0(15.5)
200	21.4(18.9)	21.0(18.6)	15.3(11.6)	20.8(18.5)	20.5(18.0)	20.1(17.5)	19.6(17.2)	18.9(16.4)	17.8(15.4)
240	21.2(18.5)	20.9(18.2)	20.9(18.4)	20.6(18.1)	20.6(17.8)	19.9(17.2)	19.4(16.9)	18.5(16.0)	17.7(15.1)
250	21.1(18.7)	20.6(18.1)	20.7(18.3)	20.6(18.1)	20.3(17.9)	19.8(17.3)	19.2(16.8)	18.5(15.9)	17.7(15.1)
300	21.1(18.4)	20.8(18.3)	20.6(18.0)	20.5(18.0)	20.3(17.5)	19.9(17.1)	19.2(16.6)	18.5(16.0)	17.5(15.1)
400	20.9(18.4)	20.5(18.0)	20.4(17.9)	20.3(17.8)	20.1(17.4)	19.6(17.1)	19.1(16.5)	18.3(15.8)	17.3(14.8)
480	20.7(18.2)	20.5(17.7)	20.4(17.7)	20.2(17.6)	20.0(17.4)	19.5(16.9)	18.9(16.5)	18.2(15.7)	17.2(14.6)
500	20.7(18.1)	20.4(17.7)	20.2(17.7)	20.0(17.4)	19.9(17.3)	19.4(16.8)	18.8(16.4)	18.0(15.4)	17.2(14.5)
600	20.6(18.1)	20.3(17.6)	20.3(17.7)	20.1(17.3)	19.9(17.4)	19.4(16.7)	18.8(16.3)	18.0(15.4)	17.1(14.3)
800	20.5(17.9)	20.2(17.5)	20.0(17.5)	20.0(17.2)	19.6(16.8)	19.2(16.6)	18.6(15.9)	17.9(15.0)	17.0(14.2)
1000	20.3(17.5)	19.9(17.1)	19.9(17.3)	19.7(17.0)	19.5(16.7)	19.0(16.1)	18.4(15.7)	17.7(15.2)	16.8(14.2)
1200	20.1(17.4)	20.0(17.2)	19.8(17.1)	19.7(17.2)	19.4(16.7)	19.0(16.3)	18.4(15.8)	17.7(14.9)	16.8(14.1)
2000	19.9(17.3)	19.6(16.8)	19.5(16.8)	19.4(16.7)	19.1(16.4)	18.8(16.2)	18.2(15.5)	17.5(14.7)	16.6(13.7)
2400	19.8(17.2)	19.6(16.8)	19.5(16.8)	19.4(16.5)	19.2(16.3)	18.8(16.0)	18.2(15.2)	17.4(14.7)	16.5(13.6)
4000	17.4(14.0)	17.4(14.6)	17.4(14.7)	17.3(14.1)	17.3(14.1)	17.2(13.7)	17.1(14.3)	16.7(13.9)	16.0(13.0)
4800	17.4(14.3)	17.4(14.5)	17.3(14.0)	17.3(14.4)	17.3(14.0)	17.2(14.0)	17.0(14.0)	16.7(13.9)	15.9(13.1)
8000	13.9(10.8)	13.9(10.5)	14.0(10.9)	13.9(10.9)	14.0(10.8)	13.9(10.5)	14.0(10.9)	13.9(10.9)	13.9(11.0)
9600	14.0(10.9)	14.0(11.0)	13.9(10.9)	14.0(11.0)	13.9(10.5)	13.9(10.5)	13.9(10.8)	13.9(10.7)	13.9(10.7)

REGISTER MAPS

There are total three 24-bit registers inside the device. These registers are used to configure and control the ADC to the desired mode of operation. These registers can be accessed through the SPI-compatible serial interface by using register read and write commands. At power-on or reset, the registers default to their initial settings, as shown in the *Reset Value* column of Table 5.

Table 5. Register map

ADDR.	NAME	RESET VALUE	BIT 23	BIT 22	BIT 21	BIT 20	BIT 19	BIT18	BIT 17	BIT 16
			BIT 15	BIT 14	BIT 13	BIT 12	BIT 11	BIT10	BIT 9	BIT 8
			BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
0x01	CONF0	0xX00000	EXTCLK	DATACHK	0	0	VBIAS	FAST_ON	0	REF_EN
			BUF	DELAY[2:0]			BCS	IDAC[2:0]		
			IDAC_PAIR	DACREF_EX	IDAC2_MUX[2:0]			IDAC1_MUX[2:0]		
0x02	CONF1	0x000201	LATENCY	SINGLE	REFSEL[1:0]		PGA[3:0]			
			FILTER	REJ60	DATA RATE[5:0]					
			MUXP[3:0]				MUXN[3:0]			
0x03	OFFSET	0x000000	OFFSET[23:16]							
			OFFSET[15:8]							
			OFFSET[7:0]							
0x04	GAIN	0x400000	GAIN[23:16]							
			GAIN[15:8]							
			GAIN[7:0]							

CONF0 Register

Table 6. CONF0 Register (Address = 0x01)

BIT 23	BIT 22	BIT 21	BIT 20	BIT 19	BIT18	BIT 17	BIT 16
BIT 15	BIT 14	BIT 13	BIT 12	BIT 11	BIT10	BIT 9	BIT 8
BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
EXTCLK	DATACHK	0	0	VBIAS	FAST_ON	0	REF_EN
BUF	DELAY[2:0]			BCS	IDAC[2:0]		
IDAC_PAIR	DACREF_EX	IDAC2_MUX[2:0]			IDAC1_MUX[2:0]		

Power-On/Reset value = 0xX00000

Bits	Bit Name	Access	Reset	Description
23	EXTCLK	R	1'bx	**Clock Source Indicator:** This bit needs to be logic 0 during register write operation, otherwise it will affect the ADC operation. 0: Internal oscillator 1: External clock
22	DATACHK	R/W	1'b0	**ADC Data Checksum Enable Bit:** 0: Disabled (default) 1: Enabled
21:20	RESERVED	R/W	2'b00	**Reserved** Always write 2'b00
19	VBIAS	R/W	1'b0	**VBIAS Enable Bit:** When this bit is enabled, AINCOM is internally connected to a voltage source at the level of middle analog supply.
18	FAST_ON	R/W	1'b0	**Fast Startup Bit:** When this bit is set to logic 1, internal reference and IDAC stays on after single shot conversion to have fast startup with fully settled data after wakeup. 0: Disabled (default) 1: Enabled
17	RESERVED	R/W	1'b0	**Reserved** Always write 1'b0

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16	REF_EN	R/W	1'b0	**Internal Reference Enable Bit:** When this bit is enabled, 2.5V internal reference referred to AVSS is enabled and outputted to REFP/REFOUT pin. Capacitor with the value of 0.1-10uF can be connected between REFP/REFOUT and AVSS pins to improve the internal reference noise. Keep in mind to short REFN pin to AVSS externally if REF_EN bit is set to logic 1, and REFP/REFN inputs are configured as ADC reference source with bits REFSEL[1:0]=2'b00 in CONF1 register.
15	BUF	R/W	1'b0	**Buffer Enable Bit:** When this bit is set to logic 1, analog input is buffered. Otherwise the input buffer is bypassed. The buffer can only be disabled when the gain equals 1 or 2. For higher gains, the buffer is automatically enabled.
14:12	DELAY[2:0]	R/W	3'b000	**Conversion Delay:** Provides additional delay from conversion start to the beginning of the actual conversion. 000: 0μs (default) 001: 104.2μs 010: 416.7μs 011: 1.667ms 100: 6.667ms 101: 26.67ms 110: 106.7ms 111: 426.7ms
11	BCS	R/W	1'b0	**Burnout Current Sources:** When this bit is enabled, a pair of 1μA current sources are applied to analog inputs to source positive input and sink negative input. 0: Disabled (default) 1: Enabled
10:8	IDAC[2:0]	R/W	3'b000	**IDAC Current Bits:** These bits select the value of the excitation current sources applied to the analog input pins. 000: Off (default) 001: 10μA 010: 50μA 011: 100μA 100: 200μA 101: 250μA 110: 500μA 111: 1000μA
7	IDAC_PAIR	R/W	1'b0	**IDAC PAIR Enable Bit:** Set this bit while both IDAC1 and IDAC2 are used as a pair to improve the IDAC matching and measurement accuracy. 0: Disabled (default) 1: Enabled
6	DACREF_EX	R/W	1'b0	**IDAC Reference Control Bit:** As default, the internal 2.5V reference is used to generate the excitation current sources. The bit also provides the option to use the voltage at VBIAS pin as the voltage source for IDAC, the magnitude of IDAC scales with the voltage applied at VBIAS pin. 0: Internal 2.5V Reference. (default) 1: External Voltage at VBIAS pin.
5:3	IDAC2_MUX[2:0]	R/W	3'b000	**IDAC2 Output Multiplexer:** Selects the analog input pin to connect IDAC2. 000: AIN0 001: AIN1 010: AIN2 011: AIN3 100: AIN4 (SIG24131/SIG24132 only) 101: AIN5 (SIG24131/SIG24132 only) 110: AIN6 (SIG24131/SIG24132 only)

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				111: AIN7 (SIG24131/SIG24132 only)
2:0	IDAC1_MUX[2:0]	R/W	3'b000	**IDAC1 Output Multiplexer:** Selects the analog input pin to connect IDAC1. 000: AIN0 001: AIN1 010: AIN2 011: AIN3 100: AIN4 (SIG24131/SIG24132 only) 101: AIN5 (SIG24131/SIG24132 only) 110: AIN6 (SIG24131/SIG24132 only) 111: AIN7 (SIG24131/SIG24132 only)

CONF1 Register

Table 7. CONF1 Register (Address = 0x02)

BIT 23	BIT 22	BIT 21	BIT 20	BIT 19	BIT18	BIT 17	BIT 16
BIT 15	BIT 14	BIT 13	BIT 12	BIT 11	BIT10	BIT 9	BIT 8
BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
LATENCY	SINGLE	REFSEL[1:0]		PGA[3:0]			
FILTER	REJ60	DATA RATE[5:0]					
MUXP[3:0]				MUXN[3:0]			

Power-On/Reset value = 0x000201

Bits	Bit Name	Access	Reset	Description
23	LATENCY	R/W	1'b0	**Zero Latency Bit:** When this bit is set, the ADC settles in one conversion cycle so that it functions as a zero latency ADC. Please see Table 12 for the actual output data rate with the combination of LATENCY and FILTER bits selection.
22	SINGLE	R/W	1'b0	**Single Shot Mode Bit:** When this bit is set, the ADC first starts one conversion and then goes to standby or sleep mode based on FAST_ON bit settling in CONF0 register. 0: Continuous Conversion (default) 1: One Shot Conversion
21:20	REFSEL[1:0]	R/W	2'b00	**Reference Selection Bits:** Select the reference source for the ADC. 00: REFP, REFN (default) 01: REFP1, REFN1 (SIG24132 only) 10: 2.5V Internal Reference 11: 1.17V Internal Reference
19:16	PGA[3:0]	R/W	4'b0000	**PGA Gain Configuration:** Selects the PGA gain. If PGA gain is set to other than 1/2, the BUF bit setting is ignored with PGA enabled. 0000: Gain=1 (default) 0001: Gain=2 0010: Gain=4 0011: Gain=8 0100: Gain=16 0101: Gain=32 0110: Gain=64 0111: Gain=128 1000: Gain=256 Others: Reserved
15	FILTER	R/W	1'b0	**Digital Filter Configuration:** Configures the ADC digital filter. 0: SINC1 filter (default) 1: SINC4 filter

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14	REJ60	R/W	1'b0	**50/60Hz Notch Bit:** This bit can only be effective while DATA RATE[5:0] =0x09 is selected. A filter notch is placed at 60Hz to allow simultaneous 50Hz/60Hz rejection. Please see Table 13 for the conversion time and 50Hz/60Hz rejection.
13:8	DATA RATE[5:0]	R/W	6'b000010	**Data Rate Configuration:** Selects the ADC data rate. 000000: 2.5SPS 000001: 5SPS 000010: 10SPS (default) 000011: 12.5SPS 000100: 15SPS 000101: 20SPS 000110: 25SPS 000111: 30SPS 001000: 40SPS 001001: 50SPS 001010: 60SPS 001011: 80SPS 001100: 100SPS 001101: 120SPS 001110: 160SPS 001111: 200SPS 010000: 240SPS 110001: 250SPS 010001: 300SPS 010010: 400SPS 010011: 480SPS 110100: 500SPS 010100: 600SPS 010101: 800SPS 110110: 1000SPS 010110: 1200SPS 110111: 2000SPS 010111: 2400SPS 111000: 4000SPS 011000: 4800SPS 111001: 8000SPS 011001: 9600SPS Others: Reserved
7:4	MUXP[3:0]	R/W	4'b0000	**Positive Input Multiplexer:** Selects the positive input multiplexer. For Internal temperature sensor and power supply measurement, buffer need to be on and PGA gain need to be ≤4 in order to get accuracy measurement. The measurement for internal temperature sensor is about 113.3mV at 27°C with sensitivity of 377.6uV/°C. 0000: AIN0 (default) 0001: AIN1 0010: AIN2 0011: AIN3 0100: AIN4 (SIG24131/SIG24132 only) 0101: AIN5 (SIG24131/SIG24132 only) 0110: AIN6 (SIG24131/SIG24132 only) 0111: AIN7 (SIG24131/SIG24132 only) 1000: AINCOM 1001: Internal temperature sensor positive. 1010: REFP 1011: AVSS 1100: DVSS 1101: (AVDD-AVSS)/4+ 1110: (DVDD-DVSS)/4+

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				Others: Reserved
3:0	MUXN[3:0]	R/W	4'b0001	**Negative Input Multiplexer:** Selects the negative input multiplexer. 0000: AIN0 0001: AIN1 (default) 0010: AIN2 0011: AIN3 0100: AIN4 (SIG24131/SIG24132 only) 0101: AIN5 (SIG24131/SIG24132 only) 0110: AIN6 (SIG24131/SIG24132 only) 0111: AIN7 (SIG24131/SIG24132 only) 1000: AINCOM 1001: Internal temperature sensor negative. 1010: REFN 1011: AVSS 1100: DVSS 1101: (AVDD-AVSS)/4- 1110: (DVDD-DVSS)/4- Others: Reserved

OFFSET Register

Table 8. OFFSET Register (Address = 0x03)

BIT 23	BIT 22	BIT 21	BIT 20	BIT 19	BIT18	BIT 17	BIT 16
BIT 15	BIT 14	BIT 13	BIT 12	BIT 11	BIT10	BIT 9	BIT 8
BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
OFFSET[23]	OFFSET[22]	OFFSET[21]	OFFSET[20]	OFFSET[19]	OFFSET[18]	OFFSET[17]	OFFSET[16]
OFFSET[15]	OFFSET[14]	OFFSET[13]	OFFSET[12]	OFFSET[11]	OFFSET[10]	OFFSET[9]	OFFSET[8]
OFFSET[7]	OFFSET[6]	OFFSET[5]	OFFSET[4]	OFFSET[3]	OFFSET[2]	OFFSET[1]	OFFSET[0]

Power-On/Reset value = 0x000000

Bits	Bit Name	Access	Reset	Description
23:0	OFFSET[23:0]	R/W	0x000000	**Offset Calibration Bits:** The 24-bit word is signed number in 2's complement format. See Calibration section for more information.

GAIN Register

Table 9. GAIN Register (Address = 0x04)

BIT 23	BIT 22	BIT 21	BIT 20	BIT 19	BIT18	BIT 17	BIT 16
BIT 15	BIT 14	BIT 13	BIT 12	BIT 11	BIT10	BIT 9	BIT 8
BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
GAIN[23]	GAIN[22]	GAIN[21]	GAIN[20]	GAIN[19]	GAIN[18]	GAIN[17]	GAIN[16]
GAIN[15]	GAIN[14]	GAIN[13]	GAIN[12]	GAIN[11]	GAIN[10]	GAIN[9]	GAIN[8]
GAIN[7]	GAIN[6]	GAIN[5]	GAIN[4]	GAIN[3]	GAIN[2]	GAIN[1]	GAIN[0]

Power-On/Reset value = 0x400000

Bits	Bit Name	Access	Reset	Description
23:0	GAIN[23:0]	R/W	0x400000	**Gain Calibration Bits:** The 24-bit word is unsigned positive number in binary format. See Calibration section for more information.

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COMMAND MAPS and PROGRAMMING

Commands are used to access the on-chip registers and also to control the operation of device. Any command starts with a valid command byte. Although register write/read and data read commands are multi-byte, consisting of command byte followed by register data, single-byte (8-bit) short commands are provided to allow fast control of device operation. The commands are listed in Table 10. Keep in mind to only send the commands listed.

The CSn input pin can be taken high between commands, or held low between consecutive commands. CSn must stay low during the entire command sequence. Pulling CSn from low to high at any time resets the SPI interface and the device is ready to accept new command.

Table 10. ADC Commands

COMMAND NAME	COMMAND BYTE								COMMAND LENGTH
	Bit 7	Bit 6	Bit 5	Bit4	Bit 3	Bit 2	Bit 1	Bit 0	
RESET	0	0	0	0	0	0	1	0	8-bit
SYNC	0	0	0	0	0	1	0	0	
SLEEP	0	0	0	0	1	0	1	0	
WAKEUP	0	0	0	0	1	1	0	0	
SYSOCAL	0	0	0	1	0	0	0	0	
SYSGCAL	0	0	0	1	0	0	0	1	
SELFOCAL	0	0	0	1	0	0	1	0	
Register Read	0	1	0	RS4	RS3	RS2	RS1	RS0	8-bit + register data
Register Write	0	0	1	RS4	RS3	RS2	RS1	RS0	8-bit + register data
Data Read	0	1	1	0	0	0	0	0	8-bit + ADC data

(1) RS4, RS3, RS2, RS1, RS0 are 5-bit register address for read/write command listed as ADDR. In Table 5.

RESET Command

The RESET command is 8-bit command. This command resets the ADC operation and resets the device registers to default.

SYNC Command

The SYNC command is 8-bit command. This command synchronizes the ADC conversion. It is very useful command if multiple ADCs exist in the system and synchronization is needed among them.

SLEEP Command

The SLEEP command is 8-bit command. This command places the ADC into low power sleep mode.

WAKEUP Command

The WAKEUP command is 8-bit command. After the WAKEUP command is received, the device exits low power sleep mode.

Calibration Commands

Three commands, SYSOCAL, SYSGCAL, and SELFOCAL, are provided to calibrate ADC offset and gain error. All the calibration commands are 8-bit command.

SYSOCAL

The SYSOCAL command initiates a system offset calibration. For a system offset calibration, the user must apply a null voltage to the AINP and AINN pins externally within the input range. The OFFSET register is updated when the command completes.

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SYSGCAL

The SYSGCAL command initiates the system gain calibration. For a system gain calibration, the user must apply a full scale voltage to the AINP and AINN pins externally within the input range. The GAIN register is updated after this operation.

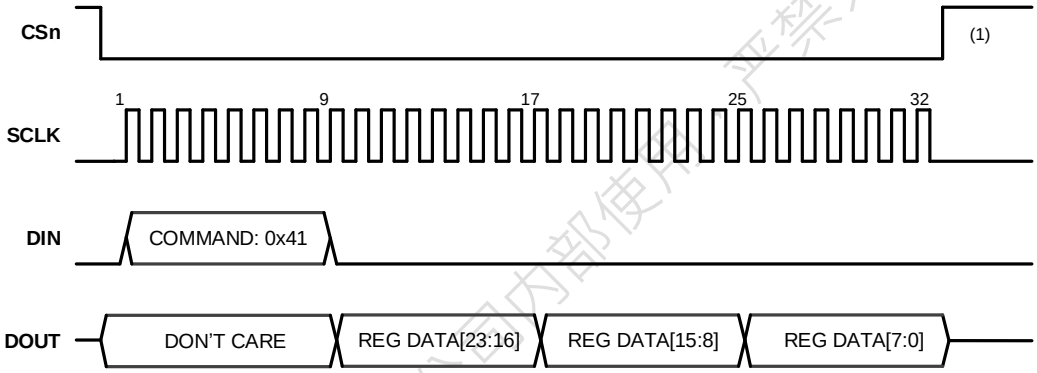
SELFOCAL

The SELFOCAL command initiates a self-offset calibration. The device disconnects both AINP and AINN inputs from ADC input and internally shorts the ADC input to mid-supply ($V_{AVDD}+V_{AVSS}$)/2 and performs the calibration. The OFFSET register is updated after this operation.

Register Read Command

The register read command is 32-bit command. The register data can only be read one at a time. Once the register read command byte is sent, the ADC responds with the corresponding register data at DOUT/DRDYn pin with most significant bit first. During the register read operation, if ADC data are ready after the command byte is sent, the loading of conversion data to output shift register are blocked to avoid data contention.

Figure 2 illustrates an example for register read operation. As shown, the command byte for reading from CONF0 register (address = 0x01) is 0x41. It is highly recommended to keep the DIN input low after the command byte is sent.



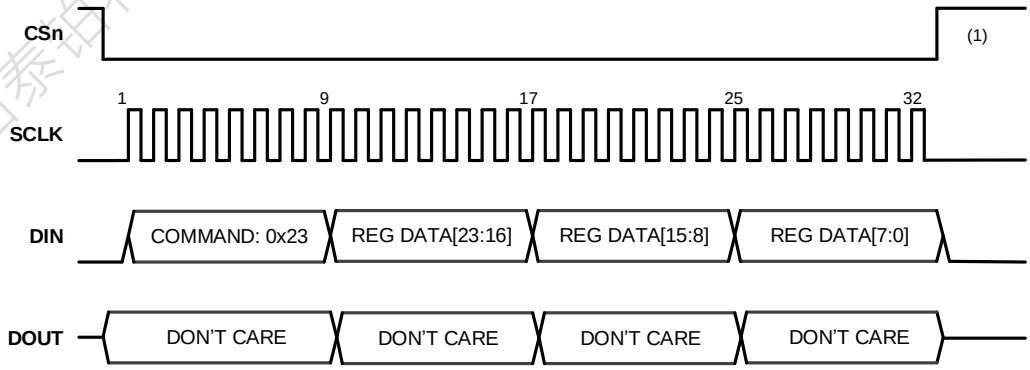
(1): Between commands, either set CSn high or keep CSn low. If CSn is kept low, the command must be completed.

Figure 2. Read Register Command Sequence

Register Write Command

The register write command is 32-bit command. The register data can only be written one at a time. Once the register write command byte for the register write command is sent, the following 24-bit is the register data to be written with most significant bit first.

Figure 3 illustrates an example for register write operation. As shown, the command byte for writing to OFFSET register (address = 0x03) is 0x23.



(1): Between commands, either set CSn high or keep CSn low. If CSn is kept low, the command must be completed.

Figure 3. Write Register Command Sequence

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Reading ADC Data

There are two ways to read the ADC conversion data: read data direct or read data by command.

Read Data Direct

As shown in Figure 4, the ADC data is 3 bytes long. Parity byte is appended to the conversion data bytes after more than 24 SCLKs is applied. Part of parity byte is formed by the checksum byte, which is the 8-bit sum of data conversion bytes plus an offset value, 0x5B. The first and last bit of parity byte are forced with 1'b1.

CHECKSUM [7:0] = DATA [23:16] + DATA [15:8] + DATA [7:0] + 0x5B

PARITY [7:0] = {1'b1, CHECKSUM [6:1], 1'b1}

After four bytes are read, the data byte sequence is repeated when more SCLKs are sent. The repeating byte sequence starts with the first byte DATA [23:16].

Make sure to keep the DIN pin low during the whole read operation. Also the read operation needs to complete 2 system clock cycles before the next new data is ready, otherwise the retrieved data is corrupted by the updating of new data on DOUT/DRDYn pin.

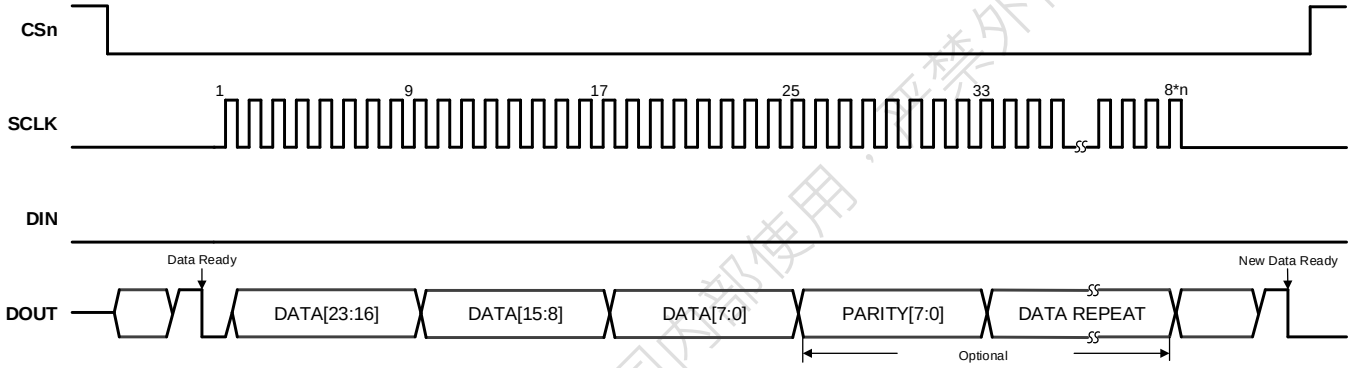


Figure 4. Read Data Direct

Read Data by Command

The data read command is 32-bit command or 40-bit command if DATACHK bit is logic 1 in CONF0 Register. The command byte for the data read command is 0x60. After the command byte is sent, the ADC responds with the 24-bit data with most significant bit first. If DATACHK bit is set to logic 1 in CONF0 Register, extra eight SCLKs are needed to retrieve the parity byte before the end of this command. Unlike read data direct method, read data by command can occur at any time without the risk of data corruption because the loading of new data to DOUT/DRDYn pin is blocked during command read operation.

Figure 5 shows the read data by command sequence. Each time after the command byte of data read command is sent, the ADC does not recognize any other command until the read operation is complete, or CSn pin is pulled high to reset SPI interface.

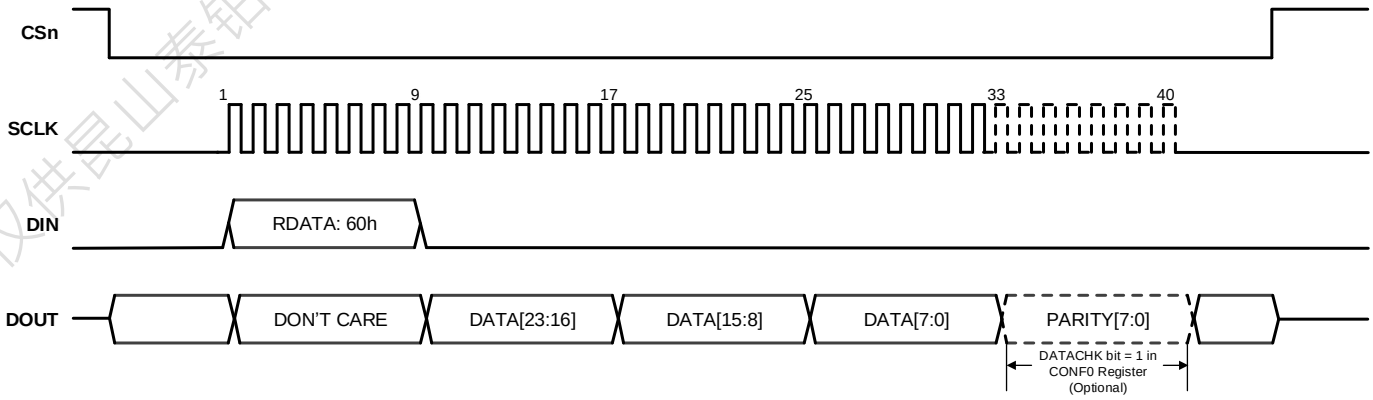


Figure 5. Read Data by Command

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CIRCUIT DESCRIPTION

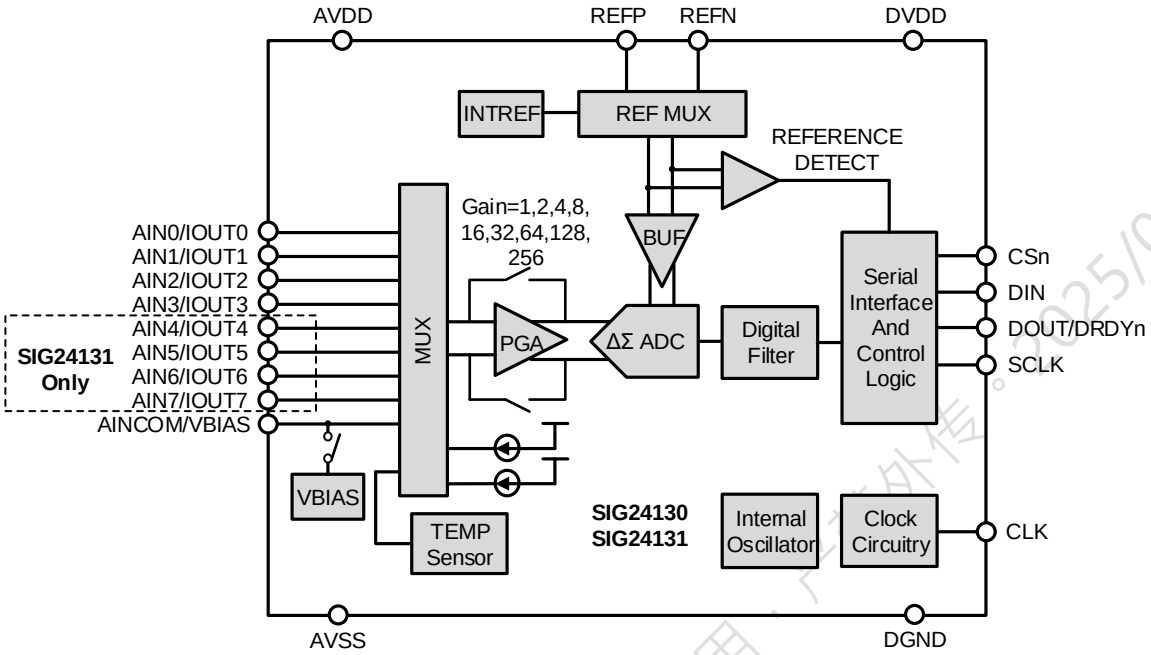


Figure 6. SIG24130/SIG24131 Block Diagram

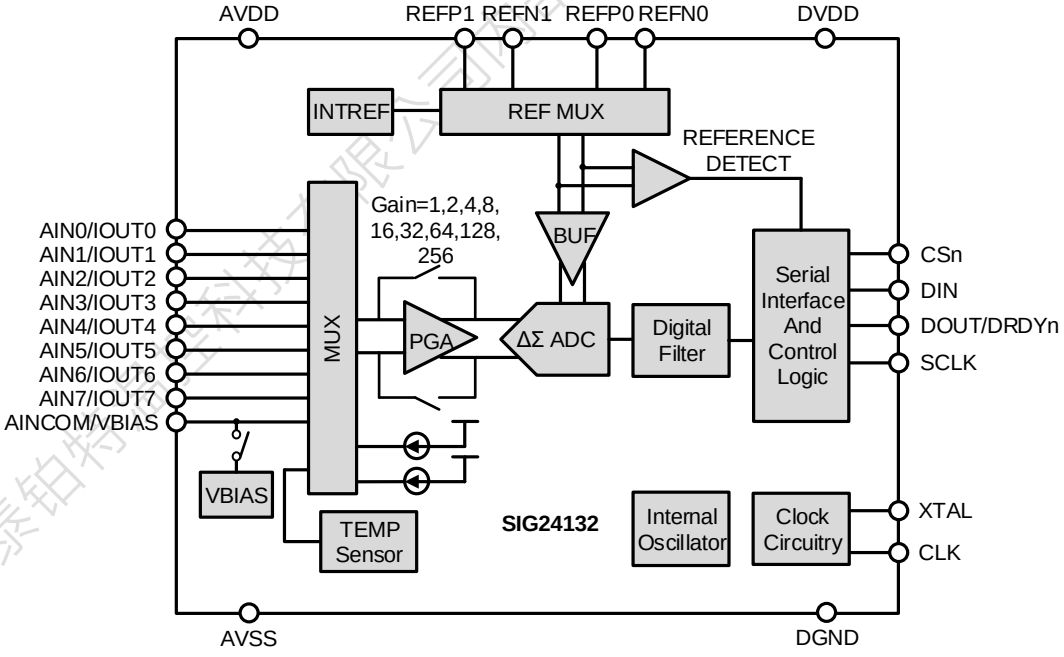


Figure 7. SIG24132 Block Diagram

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OVERVIEW

The SIG24130/1/2 are low noise, low drift, and high-resolution 24-bit analog-to-digital converters (ADC) with integrated programmable gain amplifier (PGA) and voltage reference. The ADC provides high-accuracy measurement solutions for bridge sensors, thermocouples, and resistance temperature devices (RTD). Figure 6 and Figure 7 show the device block diagram.

The ADC features a high input-impedance, low-noise, programmable gain amplifier (PGA). The PGA gain is programmable from 1, 2, 4, 8 16, 32, 64, 128, and 256 in binary steps. Also the PGA can be bypassed to allow the input range to extend 50 mV above supply or below ground. A low drift 2.5V reference is integrated on chip and two matched excitation current sources (IDACs) are provided for accurate RTD measurement. Two burnout current sources are provided at the analog inputs for sensor connection diagnosis.

A delta-sigma modulator measures the PGA output voltage according to the buffered reference voltage to provide high speed bitstream to the digital filter. Unlike SAR ADCs, this ADC is much easier to drive due to very high impedance at both analog and reference inputs.

Both SINC1 and SINC4 are provided in digital filter for the tradeoff of settling time and line-cycle rejection. Various data rates, 2.5 to 9600SPS, are provided for the tradeoff of settling time and accuracy.

Offset and gain calibration registers are provided with calibration command or direct register write to calibrate the ADC errors or overall system errors. The SPI-compatible serial interface consists of four signals: CSn, SCLK, DIN and DOUT/DRDYn. The DOUT/DRDYn pin serves as dual function of register and ADC data output and also the indicator for data ready after the conversion is done. Parity check is provided for data integrity.

The ADC has three clock options: internal oscillator, external clock, and external crystal. The nominal clock frequency is 1.536MHz.

The device can operate with bipolar ±1.35V to ±2.625V analog power supplies, or with a single 2.7V to 5.25V analog power supply. The digital power supply range is 2.7V to 5.25V.

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Programmable Gain Amplifier (PGA)

The ADC features a low-drift, low-noise, high input impedance programmable gain amplifier (PGA). The simplified diagram of the PGA is shown in Figure 8. The gain of PGA can be programmed to 1, 2, 4, 8, 16, 32, 64, 128, and 256 by setting the PGA [3:0] bits of CONF1 Register. The PGA can be bypassed to by setting the BUF bit of CONF0 Register to logic 0.

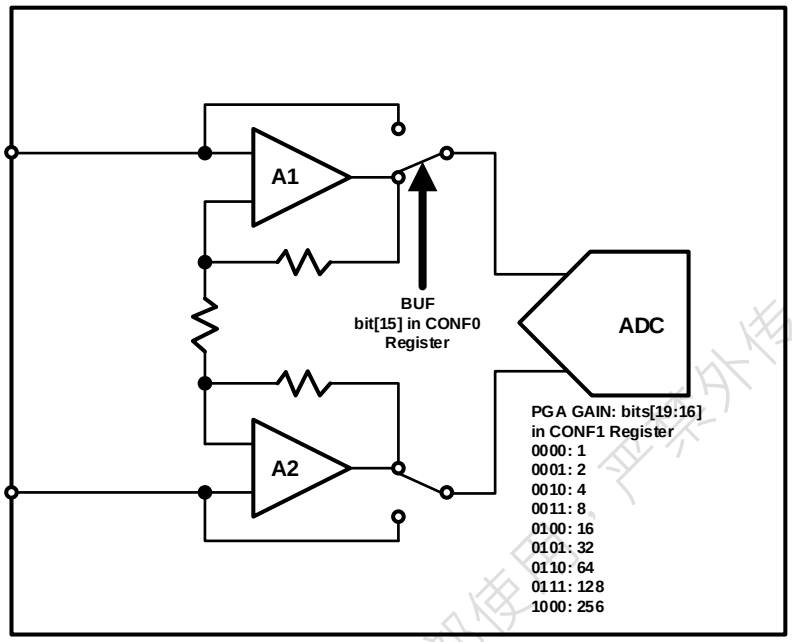


Figure 8. Simplified PGA Block Diagram

The ADC full-scale voltage range is determined by the reference voltage and the PGA gain. Table 11 shows the full-scale voltage range verses gain for 2.5V reference voltage.

Table 11. ADC Full-Scale Voltage Range with V_{REF} = 2.5V

PGA[3:0] BITS in CONF1 REGISTER	GAIN	FULL SCALE RANGE (V)(1)
0000	1	±2.500 V
0001	2	±1.250 V
0010	4	±0.625 V
0011	8	±0.312 V
0100	16	±0.156 V
0101	32	±0.078 V
0110	64	±0.039 V
0111	128	±0.0195 V
1000	256	±0.0098 V

(1) The full scale input range is proportional to V_{REF}.

Just like most amplifiers, there should be some headroom for the output of PGA to be away from the power supply (V_{AVDD}) and ground (V_{AVSS}) due to the limitation of voltage driving capability of PGA output device as shown in Figure 9. For correct linear operation, the absolute PGA output voltage must locate within the range [V_{AVSS} + 0.2, V_{AVDD} – 0.2]. The analog input common voltage must meet Equation (3):

$$(V_{AVSS} + 0.2 + V_{IN} \cdot Gain / 2) \le V_{CM} \le (V_{AVDD} - 0.2 - V_{IN} \cdot Gain / 2) \quad (3)$$

Where

V_{IN} = differential input voltage = $V_{INP} - V_{INN}$

V_{CM} = input common mode voltage = $(V_{INP} + V_{INN})/2$

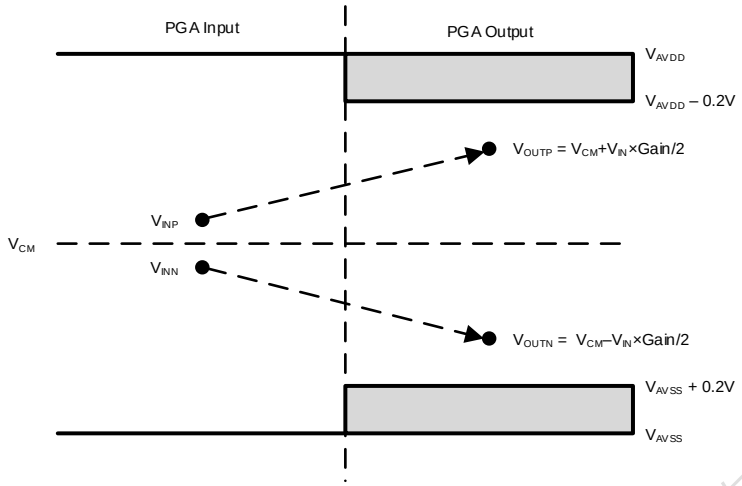


Figure 9. PGA Input and Output Range

The PGA can be bypassed by writing 0 into BUF bit in CONF1 Register. If the PGA is bypassed, the ADC absolute input voltage range extends above V_{AVDD} or below V_{AVSS}. The absolute input voltage range when the PGA is bypassed is shown in Equation (4):

$$V_{AVSS} - 0.05 < V_{INP} \text{ and } V_{INN} < V_{AVDD} + 0.05 \quad \textbf{(4)}$$

Digital Filter and Conversion Time

A delta-sigma (Σ–Δ) ADC consists of a modulator followed by a programmable digital decimation filter to produce the final high-resolution data output. Caution must be taken to choose the type of filtering based on the consideration of tradeoffs between resolution, data rate, line cycle rejection, and conversion latency.

This ADC provides two filter options: a SINC1 filter and a SINC4 filter. The filter is configured by setting the FILTER bit in CONF1 Register. When FILTER is set to 1 (default value), the SINC4 filter is selected. Otherwise the SINC1 filter is selected if FILTER is set to 0. The output data rate can be programmed from 2.5SPS to 9600SPS by DATA RATE [5:0] bits in CONF1 Register.

Figure 10 shows the frequency response of the SINC4 and SINC1 filters normalized to output data rate.

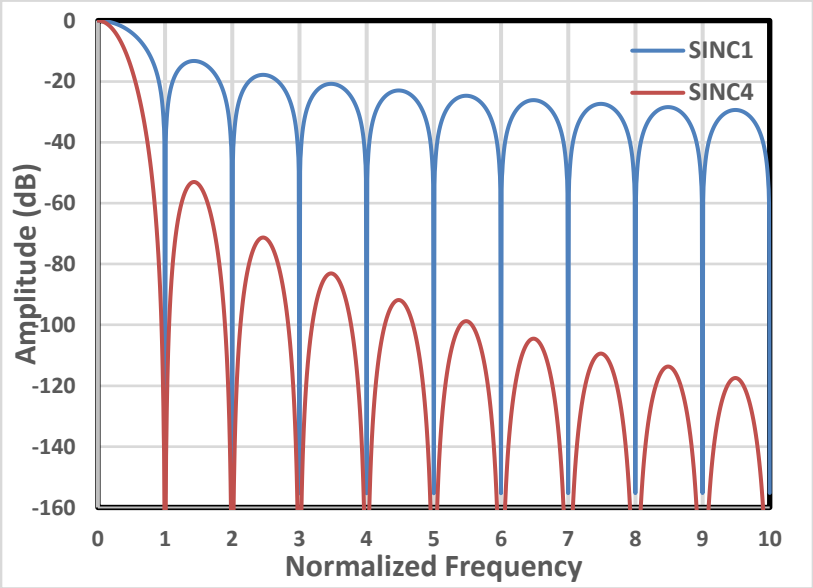


Figure 10. Frequency Response of SINC1/SINC4 Filter

While the order of the SINC filter doesn't affect the notch positions, the higher order SINC4 filter has wider notches resulting in better rejection in the band (±1 Hz) around the notches. The higher order SINC4 filter also gives better

stop-band rejection with the tradeoff of longer settling time for the same output data rate. The SINC4 filter normally takes four conversion cycles to settle, while SINC1 filter settles in one conversion cycle as shown in Figure 11.

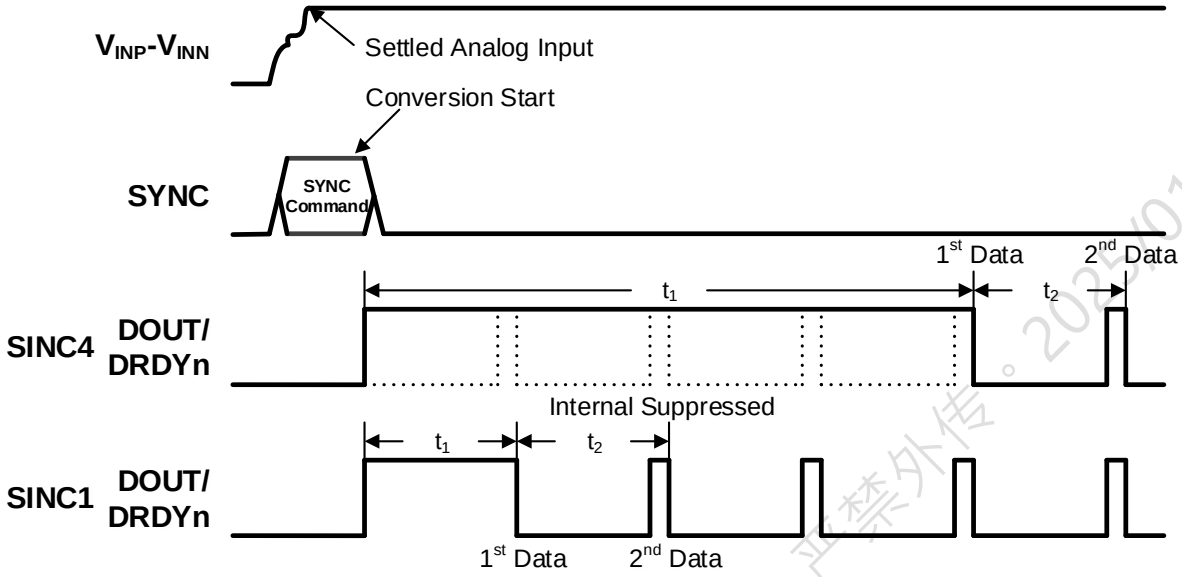


Figure 11. Conversion Sequence for SINC4 and SINC1 Filter

Table 12 gives the actual output data rate with the combination of LATENCY and FILTER bits for each data rate setting. The conversion time is the reciprocal of actual output data rate. For LATENCY=0, the table only lists the conversion time for 2nd data and thereafter, the conversion time for 1st data is the same as the conversion time with LATENCY=1 to guarantee that the first available data is always fully settled data for any LATENCY bit setting.

For the applications with fast data rate needed, where the 50-Hz and 60-Hz line-cycle rejection can't be achieved anyway, it is highly recommended to choose FILTER=0(SINC1) if faster single cycle settling is needed to measure multiple input signals with input mux switching. More options of channel switching frequency, which is listed as actual output data rate with LATENCY=1 in the table, are provided with FILTER=0(SINC1) compared to FILTER=1(SINC4). Also noise benefit can be achieved with equivalent channel switching frequency with FILTER=0(SINC1) compared to FILTER=1(SINC4).

Table 12. Output Data Rate with the Combination of LATENCY and FILTER Bits

Data Rate[5:0]	Actual Output Data Rate(SPS) (1)			
	FILTER=1(SINC4)		FILTER=0(SINC1)	
	LATENCY=0	LATENCY=1	LATENCY=0	LATENCY=1
000000	2.5	0.62	2.5	2.49
000001	5	1.25	5	4.97
000010	10	2.49	10	9.88
000011	12.5	3.11	12.5	12.3
000100	15	3.73	15	14.7
000101	20	4.96	20	19.5
000110	25	6.19	25	24.2
000111	30	7.41	30	28.9
001000	40	9.84	40	38.1
001001(REJ60=0)	50	12.2	50	47.1
001001(REJ60=1)	50	12.2	24.5	16.2
001010	60	14.6	60	55.8

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001011	80	19.4	80	72.7
001100	100	24.0	100	88.9
001101	120	28.6	120	104
001110	160	37.5	160	133
001111	200	46.2	200	160
010000	240	54.5	240	185
110001	250	55.6	250	182
010001	300	66.7	300	218
010010	400	85.7	400	267
010011	480	100	480	300
110100	500	100	500	286
010100	600	120	600	343
010101	800	150	800	400
110110	1000	167	1000	400
010110	1200	200	1200	480
110111	2000	500	2000	500
010111	2400	600	2400	600
111000	4000	1000	4000	1000
011000	4800	1200	4800	1200
111001	8000	2000	8000	2000
011001	9600	2400	9600	2400

(1) Valid for the internal oscillator or an external 1.536 MHz clock. Scales proportional with f_{CLK}. The conversion time is the reciprocal of actual output data rate and does not include the programmable delay set by the DELAY [2:0] bits in the CONF0 register.

Table 13 summarizes the 50-Hz and 60-Hz line-cycle rejection based on ±1Hz and ±2Hz variation of power-line assuming ideal system clock. Compared to SINC1 filter, the high-order SINC4 filter leads to better 50 Hz and 60 Hz rejection. This accuracy of system clock also needs to factor in during the calculation of the 50-Hz and 60-Hz line-cycle rejection while internal oscillator is used. Definitely more accurate external clock yields better line-cycle rejection. The fastest output data rate with both 50-Hz and 60-Hz line-cycle rejection supported is 16.22SPS with single cycle settling.

Table 13. 50Hz and 60Hz Line Cycle Rejection for SINC1 and SINC4 Filter

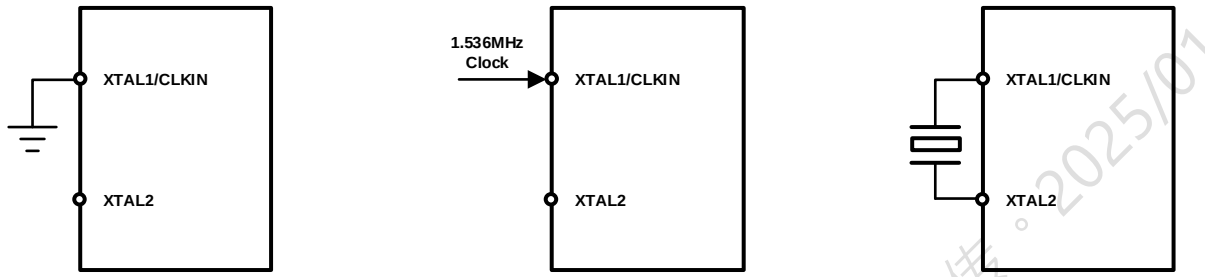
Data Rate[5:0] (SPS) (1)	50 Hz ± 1 Hz		60 Hz ± 1 Hz		50 Hz ± 2 Hz		60 Hz ± 2 Hz	
	SINC1	SINC4	SINC1	SINC4	SINC1	SINC4	SINC1	SINC4
10	-33.9	-136	-35.6	-142	-28.2	-113	-29.8	-129
20	-17.8	-71.4	-35.5	-142	-18.0	-71.9	-29.4	-118
50	-33.8	-135	-16.8	-62.3	-27.6	-110	-17.6	-60.4
50(REJ60=1)	-81.1	-115	-66.9	-82.5	-67.9	-95.5	-60.1	-75.2

(1) Valid for the internal oscillator or an external 1.536 MHz clock. Scales proportional with f_{CLK}.

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Clock Mode

The on-chip oscillator, an external clock, or an external crystal (SIG24132 only) can be used as the clock source to the device. Figure 12 illustrates the configuration for each clock mode. If the CLKIN pin is shorted to analog ground (AVSS), the internal oscillator is enabled. If an external clock is detected at the CLKIN pin, the ADC automatically selects the external clock. The clock mode can be verified by reading the EXTCLK bit, bit 23 of CONF0 Register, with value 0 as the indicator for internal oscillator.



Option A: Internal Oscillator | Option B: External Clock | Option C: Crystal Oscillator

Figure 12. Clock Mode Configurations

Power-On Reset (POR)

The ADC has two power supplies, analog and digital. The analog power supply (AVDD, AVSS) can be bipolar ±1.35V to ±2.625V or a single 2.7V to 5.25V, and is independent of the digital power supply. The digital supply (DVDD) range is 2.7V to 5V.

Figure 13 shows the POR sequence. The internal POR circuitry forces the ADC in reset state if the digital supply voltage (V_{DVDD}) is below POR voltage threshold which is about 1.3V. After the digital supply voltage (V_{DVDD}) exceeds POR voltage threshold, additional 43ms of waiting time is needed for power supply to be fully settled before sending any command, otherwise the command is ignored.

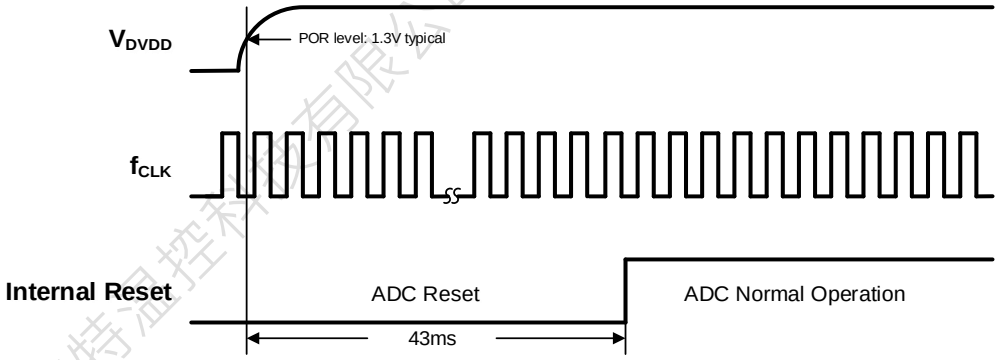


Figure 13. Power-On Reset Sequence

Burnout Current Sources

To help diagnosis a possible sensor malfunction, the device provides internal 1μA, burnout current sources. When enabled by setting the respective BCS bit in CONF0 Register, one current source sources current to one side of the analog inputs and the other current source sinks current from the other side of analog inputs.

In case of an open circuit in the sensor, these burnout current sources pull one side of the inputs towards AVDD and the other side of the inputs towards AVSS, resulting in a positive full-scale reading.

Temperature Sensor

Temperature sensor is enabled when MUXP[3:0] and MUXN[3:0] bit in CONF1 Register are both set to 4'b1001. Buffer needs to be on and PGA gain needs to be less than or equal to 4 for accurate temperature sensor measurement. Equation (5) shows how to convert the temperature sensor voltage reading to degrees Celsius (°C):

$$\text{Temperature (°C)} = (\text{Voltage Reading (uV)} - 113{,}300) / 377.6\ (\mu\text{V/°C}) + 27°\text{C} \quad \textbf{(5)}$$

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SPI Interface

The ADC provides a 3-wire or 4-wire SPI-compatible interface with SPI mode 1 supported. Please see Timing Requirements: Serial Interface section for the timing information related to serial interface.

CHIP SELECT (CSn)

The CSn pin is an active low input signal. When CSn is high, the serial interface is in reset state, any command from DIN pin is ignored, and the output DOUT/DRDYn pin is disabled as tri-state. CSn must stay low during the entire command or data retrieval operation. In case that interface synchronization is lost, pulling CSn high resets the digital interface to abort any current operation. Due to the concern of noisy environment in real application, it is highly recommended to use 4-wire SPI configuration with CSn toggling during each command or data retrieval operation.

SERIAL CLOCK (SCLK)

The serial clock is a Schmitt-triggered input to make it noise immune. This pin is used to clock data into and out of the device. Input data from DIN pin are latched on the falling edge of SCLK and output data on DOUT pin are updated on the rising edge of SCLK.

DATA INPUT (DIN)

The serial data input pin (DIN) is used to send data to the device. The device latches data from DIN pin on the falling edge of SCLK.

DATA OUTPUT (DOUT/DRDYn)

The DOUT/DRDYn pin is a dual-function output. This pin serves as the serial interface data output and also as an indicator for new data ready for retrieval. First, conversion or register data are shifted out on DOUT/DRDYn pin on the rising edge of SCLK. Second, while the SPI interface is at idle state, the DOUT/DRDYn pin goes low to indicate that new conversion data are ready for retrieval.

Data Format

The device provides 24 bits of conversion data output in binary 2's complement format, left justified, MSB first. The ADC input is bipolar-differential and is scaled such that zero differential input results in an ideal code of 0x000000, positive full scale input results in an ideal code of 0x7FFFFF, and negative full scale input results in an ideal code of 0x800000. The output clips if the signal exceeds full-scale. Table 14 lists the ideal output codes for different input signals.

Table 14. Ideal Output Code vs. Input Signal

INPUT SIGNAL VIN = VINP – VINN	IDEAL OUTPUT CODE
≥ FS (2^{23} – 1) / 2^{23}	0x7FFFFF
FS / 2^{23}	0x000001
0	0x000000
-FS / 2^{23}	0xFFFFFF
≤ –FS	0x800000

Calibration

24-bit offset and gain calibration registers are provided to calibrate the ADC errors or overall system errors. There are two ways to calibrate the ADC: issuing calibration commands to the device, or directly writing the calibration values to the calibration registers. The ADC provides three calibration commands: self-offset calibration command (SELFOCAL), system offset calibration command (SYSOCAL), and system gain calibration command (SYSGCAL).

The output of digital filter is first subtracted by the offset register OFFSET [23:0] and then multiplied by the gain register GAIN [23:0]. In order to achieve correct calibration results, it is required to calibrate the offset first before system gain calibration. Also keep in mind that the power supplies and reference voltage need to be fully settled before calibration commands are issued. Ideally the final data output can be expressed in Equation (6). And the scaling examples of offset and gain register to the final data output are listed in Table 15 and Table 16 respectively.

$$\text{Final Data Output} = \left[\frac{V_{IN} \times \text{PGA_GAIN}}{V_{REF}} \times 2^{23} - \text{OFFSET}[23:0]\right] \times \frac{\text{GAIN}[23:0]}{0x400000} \quad \textbf{(6)}$$

Table 15. Offset Calibration Register Scaling

OFFSET REGISTER VALUE	OFFSET SUBSTRACTED
0x3FFFFF	FS/2
0x1FFFFF	FS/4
0x000000	0
0xE00000	-FS/4
0xC00000	-FS/2

Table 16. Gain Calibration Register Scaling

GAIN REGISTER VALUE	GAIN FACTOR
0x100000	0.25
0x200000	0.50
0x400000	1.00
0x600000	1.50
0x800000	2.00

The DOUT/DRDYn pin goes high right after a calibration command is issued to indicate that the ADC is busy and returns low for 32 SYCLKs after the calibration is complete with calibration register updated. Then the device starts a new conversion automatically and DOUT/DRDYn goes low once a new ADC data is ready for retrieval as shown in Figure 14.

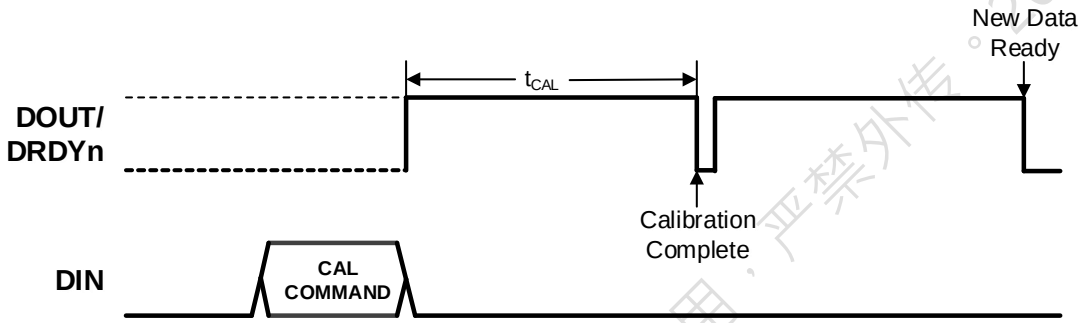


Figure 14. Calibration Timing Sequence

The calibration time (t_{CAL}) is about the same time as the normal conversion time for first data as shown in Table 12. The errors can be reduced to within the ADC noise floor by calibration. The calibration time can be shortened by temporarily changing to a faster data rate with the tradeoff of calibration accuracy. Changing the data rate or the digital filter does not require re-calibration as long as PGA gain is the same.

ADC Normal Conversion

Restart New Conversion

There are two different methods to trigger the reset of digital filter and initiate a new ADC conversion: write to CONF0 or CONF1 registers; send SYNC command. The DOUT/DRDYn pin goes high to indicate that the ADC is busy and returns low after the conversion is complete with new data ready for read out. The new data is always fully settled data as long as the selected analog input signal is stable before restarting ADC conversion. Extra delay time can be internally added before the digital filter starts to allow analog input signal to be fully stable by configuring the DELAY[2:0] bits in CONF0 Register.

Single Shot Mode

When the SINGLE bit in CONF1 Register is set to logic 1, the ADC is working as single shot mode. It first starts one conversion and then goes to standby or sleep mode based on FAST_ON bit setting in CONF0 Register. When FAST_ON bit is set to logic 1, internal reference and IDAC stays on after single shot conversion to have fast startup with fully settled data after wakeup. The ADC automatically restarts another single shot conversion by writing to CONF0 or CONF1 registers or sending SYNC command.

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REVISION HISTORY

The revision history provided is for informational purposes only and is believed to be accurate, but not warranted. Please go to web to make sure you have the latest revision.

DATE	REVISION	CHANGE
July 12, 2022		Initial release.

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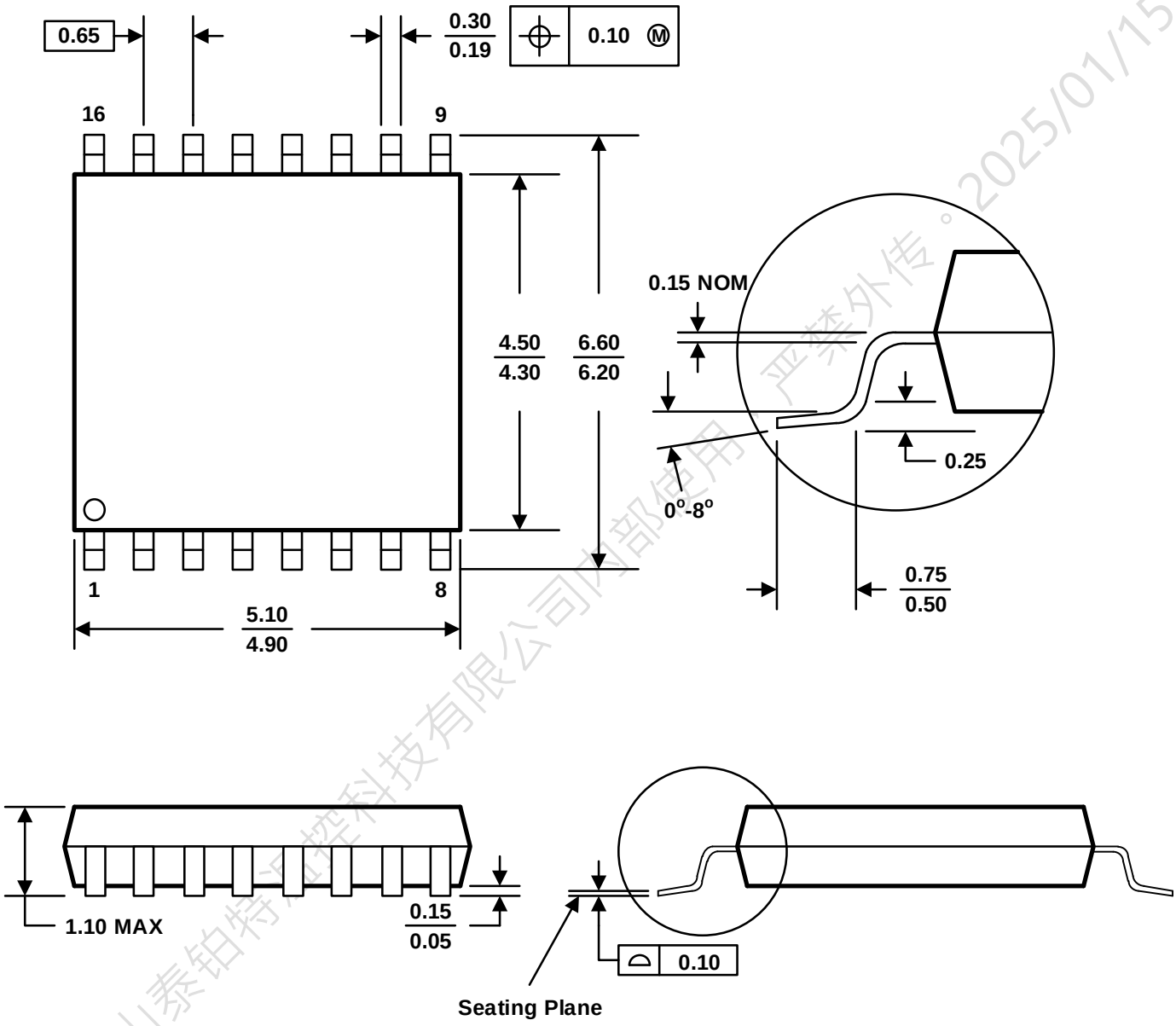
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PACKAGE OUTLINE DIMENSIONS

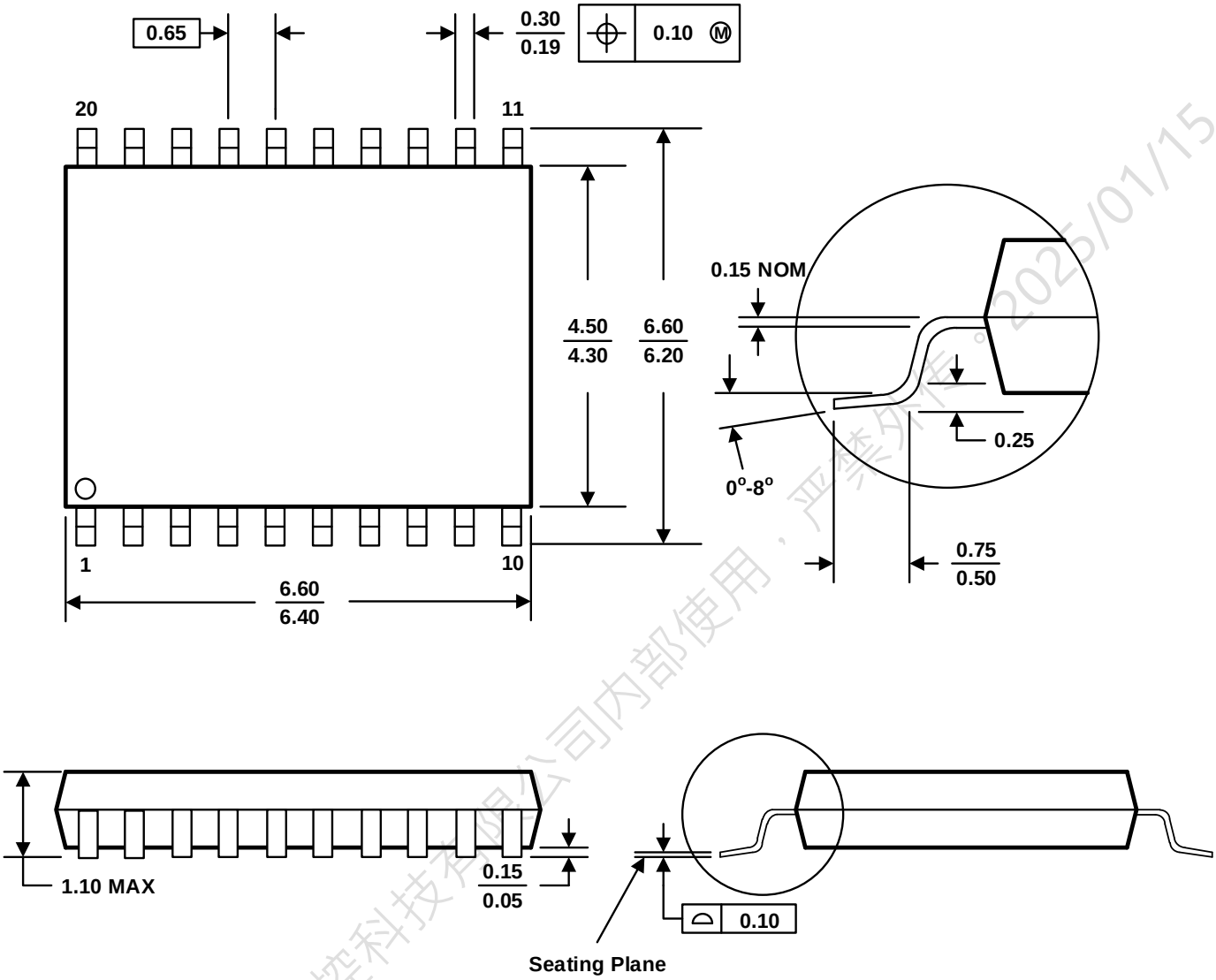
TSSOP-16



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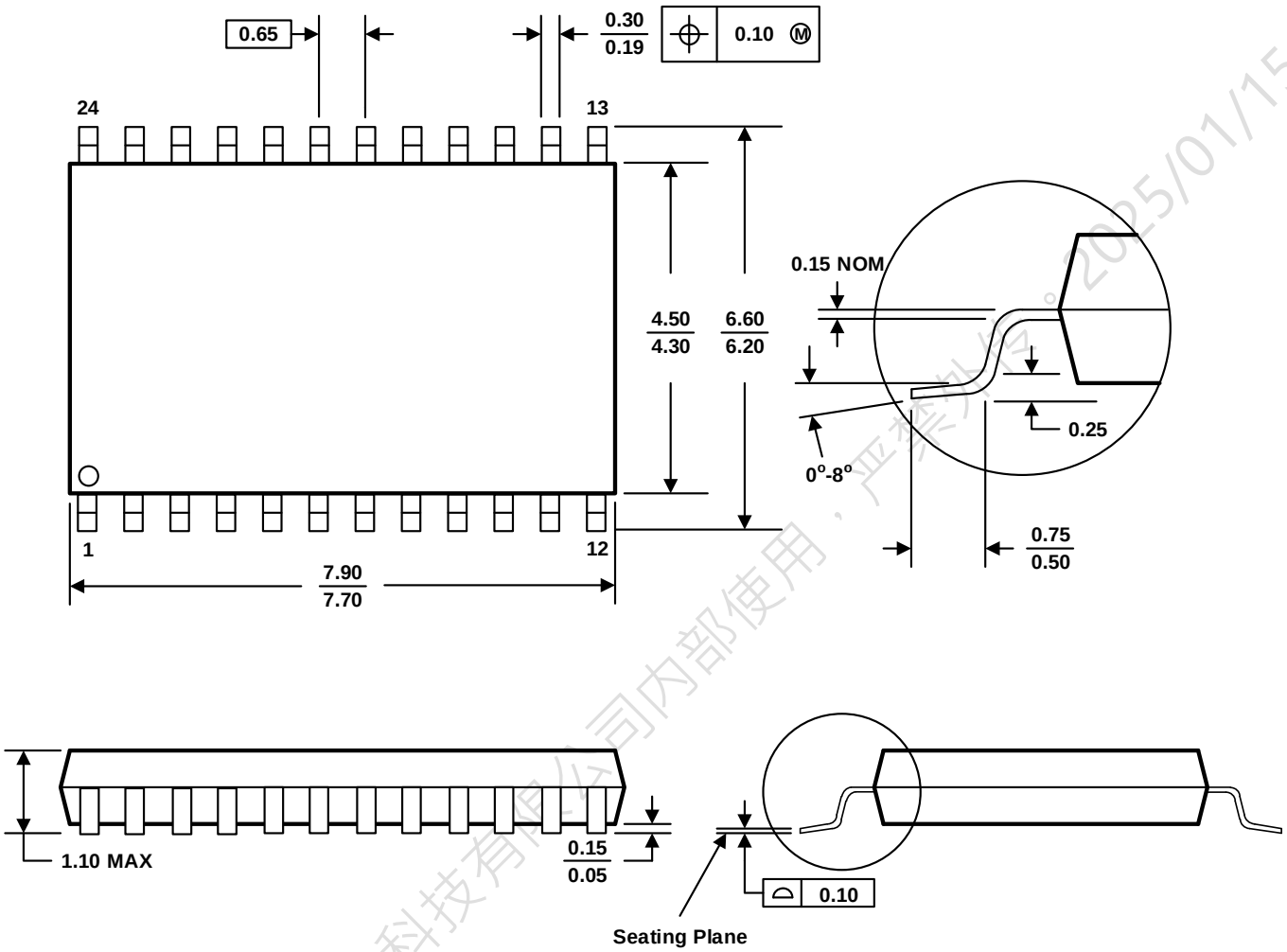
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